National Technical University of Athens
School of Mechanical Engineering
Interdepartmental Postgraduate Studies Program
Automation Systems
Direction of Automatic Control and Robotic Systems

Optimal Distributed Control of Heterogeneous Multi-Agent Systems with Non-Convex Cost Functions and Sampled Adjacent Information

Master Diploma Thesis

George Krommydas

Supervisor: Haris E. Psillakis
Assistant Professor
NTUA

Athens, October 2025

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Approved by the examination committee on 29th October 2025.

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Acknowledgements

First and foremost, I would like to acknowledge my supervisor Assistant Professor Harris E. Psillakis for his guidance, knowledge and advice, which carried me through the completion of this dissertation. I would also like to thank the other members of the committee, Associate Professors Costas Tzafestas and Charalampos Bechlioulis. I would also like to thank my family and friends for their moral support during the writing of this dissertation.

Athens, October 2025

George Krommydas

Table of Contents

List of Figures **vii**

List of Tables **ix**

List of Algorithms **xi**

Abstract **xiii**

Περίληψη **xv**

1 Introduction **1**

1.1 Motivation and Contribution 3

1.2 Literature Review 4

1.2.1 Optimal Consensus Problem 4

1.2.2 Convex and Non-Convex Optimization 4

1.2.3 Graph Topology Variations 6

1.3 Thesis Structure 7

2 Mathematical Background **9**

2.1 Graph Theory 9

2.1.1 Basic Definitions 9

2.1.2 Graph Matrices 10

2.2 Matrix Theory 11

2.3 Mathematical Analysis 12

2.4 Lyapunov Stability Theory 14

2.5 Prox-GPDA 16

3 Optimal Consensus via NOCGPROX Variables **21**

3.1 Optimal Consensus Problem 21

3.1.1 Definition of NOCGPROX Variables 21

3.2 Convergence Analysis 24

4 Distributed Control System Design for NOCGPROX Regulation 37
4.1 Heterogeneous Multi-Agent Dynamics 37
4.2 Distributed Control System Design for NOCGPROX Variables 39

5 Simulations and Results 49
5.1 Application 49
5.2 System Simulation with MATLAB 51
5.2.1 Parameters of the System 51
5.2.2 Scenario 1: Optimal Consensus with Obstacle Avoidance 53
5.2.3 Scenario 2: Formation Control with Obstacle Avoidance 56

6 Epilogue 61
6.1 Conclusions 61
6.2 Future Work 62

Bibliography 63

List of Figures

1.1 A flock of birds. 1
1.2 A swarm of fish. 1
1.3 A swarm of ants constructing a bridge. 1
1.4 Architecture of a distributed network. 2

2.1 Undirected graph topology. 9
2.2 Directed graph topology. 9
2.3 Undirected weighted graph topology. 10
2.4 Directed weighted graph topology. 10

5.1 Parameters of $i-$th UAV, [68]. 50
5.2 Communication Graph of UAVs. 51
5.3 Output trajectory x_{i1} of UAVs with respect to time. 54
5.4 Output trajectory x_{i2} of UAVs with respect to time. 54
5.5 Output trajectory x_{i3} of UAVs with respect to time. 54
5.6 Control input u_{i1} of UAVs with respect to time. 55
5.7 Control input u_{i2} of UAVs with respect to time. 55
5.8 Control input u_{i3} of UAVs with respect to time. 55
5.9 NOCGPROX variables $\|q_i\|$ with respect to time. 56
5.10 Formation control of the agents. 57
5.11 Output trajectory x_{i1} of UAVs with respect to time. 58
5.12 Output trajectory x_{i2} of UAVs with respect to time. 58
5.13 Output trajectory x_{i3} of UAVs with respect to time. 58
5.14 Control input u_{i1} of UAVs with respect to time. 59
5.15 Control input u_{i2} of UAVs with respect to time. 59
5.16 Control input u_{i3} of UAVs with respect to time. 59
5.17 NOCGPROX variables $\|q_i\|$ with respect to time. 59

List of Tables

5.1 Mass of every agent. 52
5.2 Variables of simulation 53

List of Algorithms

1 The Proximal Gradient Primal Dual Algorithm (Prox-GPDA) 18

Abstract

Multi-agent systems inherently face synchronization and communication constraints, which make their control particularly challenging. Within this framework, the optimal consensus problem is a key topic of interest. In this problem, each agent is assigned a cost function dependent on its output vector, and information is exchanged among neighboring agents according to the communication graph topology. The objective is to drive all agents' outputs to a common value that minimizes the sum of these cost functions.

This thesis investigates the optimal consensus problem for systems with non-convex cost functions and connected communication topologies. Considering bandwidth limitations, agents exchange position vectors at discrete sampling intervals. A regulation algorithm based on NOCGPROX variables, representing the agents' position errors, is proposed. Taking into account the individual dynamics of each agent, distributed robust control laws are designed to ensure convergence of the outputs to the optimal point. These laws combine the sliding surface method, to achieve robustness, with feedback linearization, to compensate for system nonlinearities.

Theoretical analysis shows that the proposed control laws drive the error variables and their squared integrals to zero, ensuring convergence to an optimal consensus point. The algorithm and control laws were validated through simulations on a swarm of UAVs, demonstrating accurate and reliable performance in MATLAB.

Keywords

Optimal Consensus, Distributed Non-Convex Optimization, Multi-Agent Systems, Asynchronous Communication, Distributed Automatic Control.

Περίληψη

Τα πολυπρακτορικά συστήματα από τη φύση τους εμφανίζουν προβλήματα συγχρονισμού και επικοινωνιών που καθιστούν δύσκολο τον έλεγχό τους. Στο πλαίσιο αυτό, το πρόβλημα της βέλτιστης συναίνεσης αποτελεί ένα κύριο θέμα ενδιαφέροντος. Σε αυτό το πρόβλημα, ανατίθεται σε κάθε πράκτορα μία τοπική συνάρτηση κόστους που εξαρτάται από το διάνυσμα εξόδου του, με την πληροφορία να ανταλλάσσεται μεταξύ των γειτονικών πρακτόρων σύμφωνα με την τοπολογία του γράφου επικοινωνίας. Ο στόχος είναι να οδηγηθούν οι έξοδοι των πρακτόρων σε μια κοινή τιμή, η οποία ελαχιστοποιεί το άθροισμα των συναρτήσεων κόστους.

Η παρούσα διατριβή διερευνά το πρόβλημα της βέλτιστης συναίνεσης για συστήματα με μη κυρτές συναρτήσεις κόστους και συνδεδεμένες τοπολογίες γράφων. Λαμβάνοντας υπόψιν τους περιορισμούς του εύρους ζώνης επικοινωνιών, οι πράκτορες ανταλλάσουν το διάνυσμα εξόδου σε διακριτά διαστήματα δειγματοληψίας. Προτείνεται ένας αλγόριθμος ρύθμισης βασισμένος σε μεταβλητές NOCGPROX, που αντιπροσωπεύουν τα σφάλματα θέσης των πρακτόρων. Λαμβάνοντας υπόψιν τη δυναμική του εκάστοτε πράκτορα, σχεδιάστηκαν εύρωστοι κατανεμημένοι νόμοι ελέγχου οι οποίοι εξασφαλίζουν σύγκλιση των εξόδων στο βέλτιστο σημείο. Αυτοί οι νόμοι συνδυάζουν την μέθοδο της επιφάνειας ολίσθησης, για να επιτευχθεί ευρωστία, μαζί με γραμμικοποίηση με ανάδραση, για να αντισταθμιστούν οι μη γραμμικοί όροι της δυναμικής του συστήματος.

Η θεωρητική ανάλυση δείχνει πως οι προτεινόμενοι νόμοι ελέγχου οδηγούν τις μεταβλητές σφάλματος στο μηδέν, εξασφαλίζοντας τη σύγκλιση σε ένα βέλτιστο σημείο συναίνεσης. Ο αλγόριθμος και οι νόμοι ελέγχου επικυρώθηκαν μέσω προσομοιώσεων σε ένα σμήνος από UAVs, αποδεικνύοντας ακριβή και αξιόπιστη απόδοση στο MATLAB.

Λέξεις Κλειδιά

Βέλτιστη Συναίνεση, Μη Κυρτή Κατανεμημένη Βελτιστοποίηση, Πολυπρακτορικά Συστήματα, Ασύγχρονες Επικοινωνίες, Κατανεμημένος Αυτόματος Έλεγχος.

Chapter 1

Introduction

In recent years, there has been an enormous growth in the field of multi-agent systems. More and more complicated systems have been designed, in order to mimic the motion of animals, which operate in swarms, such as birds, fish, and insects (Figures 1.1–1.3). Due to their inherent complexity, new analytical approaches have been introduced to study and control such systems effectively. One of the most influential ideas in this context is the decomposition of a large system into interconnected subsystems, which communicate through properly defined links. These interconnected subsystems form what is known as a distributed system.

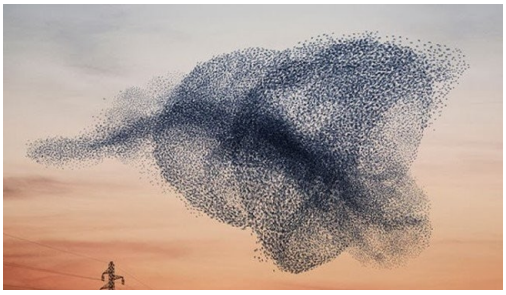

Figure 1.1: A flock of birds.

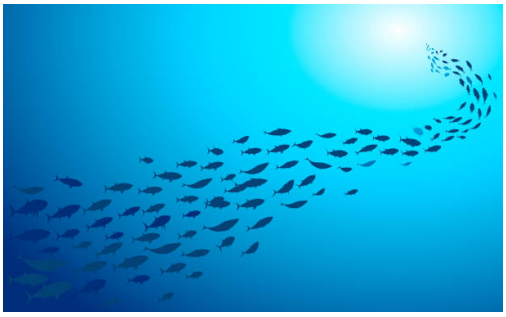

Figure 1.2: A swarm of fish.

Figure 1.3: A swarm of ants constructing a bridge.

A distributed system is typically represented as a graph-based architecture, where the nodes correspond to agents and the edges represent communication channels. Through these channels, each agent exchanges information with its neighbors in a distributed manner (Figure 1.4). One of the fundamental problems studied within this framework is the consensus problem, where agents aim to reach an agreement on certain state variables despite limited and local information exchange. According to [1], numerous consensus protocols have been developed to reproduce the coordination and perception observed in natural swarms. In these systems, agents exhibit swarm-like behavior, collectively achieving a shared objective.

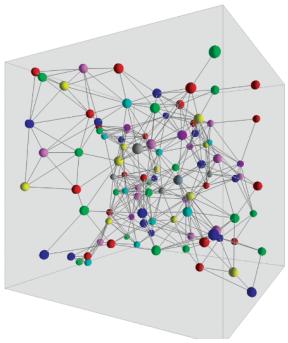

Figure 1.4: Architecture of a distributed network.

To achieve consensus, distributed systems rely on distributed control algorithms, which generate commands that guide each agent's actions [2],[3]. The study of distributed control has attracted significant attention, leading to the development of various control methodologies. Two main approaches have emerged: centralized control and decentralized (or distributed) control. In centralized control, a single master agent (or central controller) processes all information and provides commands to subordinate agents, which follow the instructions. In contrast, distributed control allows each agent to operate autonomously using locally available information, while maintaining communication with its neighbors. Although distributed control is inherently more complex, it is also more practical for real-world applications, as physical systems are subject to limitations such as finite communication bandwidth and restricted communication ranges.

As multi-agent systems increase in scale and complexity, control objectives often extend beyond achieving consensus to reaching optimal coordination among agents. In such cases, the goal is not only for the agents to agree on a common value but also to ensure that this common value minimizes a global performance criterion. This requirement naturally introduces the framework of distributed optimization.

In distributed optimization, each agent is assigned a local cost function that represents its individual performance or constraints. Through communication with neighboring agents, the network collectively minimizes the sum of these cost functions, subject to the communication and dynamic limitations of the system. This formulation captures many practical coordination problems in engineering, where agents must cooperatively optimize shared objectives under limited information exchange [2].

A particularly important case within this framework is the optimal consensus problem, in which agents seek to reach agreement on a common output that corresponds to the global minimum of the aggregate cost function. This problem unifies consensus dynamics and optimization principles, providing a foundation for modern cooperative control design. By integrating methods from distributed optimization and robust control theory, researchers have developed algorithms capable of ensuring convergence and resilience against disturbances, modeling uncertainties, and communication constraints.

The study of multi-agent systems has therefore evolved toward the synthesis of distributed control and distributed optimization techniques. This integration enables the design of scalable, robust, and efficient coordination strategies that can address complex real-world challenges in autonomous systems, sensor networks, and cooperative robotics.

1.1 Motivation and Contribution

The aim of this thesis is to solve the optimal consensus problem, with non-convex constraints, for quadcopters who exchange partial information of their position with their neighbors, obtained at specific sampling instants. In addition, the algorithm will lead the agents in the region of the optimal point. In order to achieve this goal, NOn-Convex Gradient PROXimal (NOCGPROX) variables are defined, which are added into the Prox-GPDA algorithm, [4], to prove the proximal optimal consensus. The development of robust distributed control laws that guarantee that the NOCGPROX variables are $\mathcal{L}_1$ is a crucial prerequisite for the convergence of the algorithm. In this manner, optimal consensus will be achieved. Additionally, the quadcopters will be driven to the optimal consensus point. Furthermore, the signals of the closed loop system have to be bounded.

The thesis' contribution can be summarized as follows.

- The NOCGPROX variables are introduced, that recast the optimal consensus problem into a regulation one. This allows the problem to be solved with distributed control techniques, while achieving proximal optimal consensus, for agents with nonlinear dynamics.

- The proposed method for solving the problem depends on sampled filtered output measurements of the agents, thus achieving low frequencies in communication between them. Furthermore, it requires only the knowledge of the Laplacian matrix and the gradient of the local cost functions. In order to have stability in the system, it requires robust distributed control laws which ensures square integration of the variables. This improves the outline of the algorithm in cases in which the optimal solutions are not inside a feasible set.

- The design parameters of the algorithm must have large values to ensure sublinear stability of the method. In addition, NOCGPROX variables must tend to zero, to

achieve optimal consensus.

1.2 Literature Review

1.2.1 Optimal Consensus Problem

A lot of scientific studies have been conducted in the years regarding distributed optimization. The foundation of distributed optimization was demonstrated in [5], in which a variety of asynchronous deterministic and stochastic methods have been developed with the use of gradient. From that point on, network systems have been designed and researched, putting into use distributed optimization algorithms for searching the best solution possible, either in discrete or continuous time.

Regarding the application of distributed optimization in the consensus problem, most researchers have made use of discrete time algorithms in an attempt to reduce computational costs. The concept of the consensus problem has been studied throughout the years, with many scientific papers being published. An interest approach is presented in [6]. The consensus between the agents is achieved through an adaptive schema which guarantee boundedness and regulation of the proposed variables with the sampled measurements being delayed. Additionally, in [7] and [8] is studied the optimal consensus problem. The optimal consensus is being achieved by utilizing a nonlinear PI controller, [9], which regulates the proposed variables with few transmissions of output measurements. In [10] another solution is proposed, solving a proximal aspect of the optimal consensus problem. The solution is based on EXTRA algorithm for optimization, [11]. The proposed variables ensure approximate consensus to a neighborhood of zero.

1.2.2 Convex and Non-Convex Optimization

There have been published numerous proposition for optimization algorithms in multi-agent systems. An algorithm whose being studied a lot is Alternating Direction Method of Multipliers (ADMM), [12]. In [13] has been proposed a different approach to the method. A more relaxed version is being developed which does not require synchronous updates between the agents with the constraints being convex, and achieve linear stability and convergence. On the other hand, in [14] is proposed a proximal form of the algorithm for non-convex and non-smooth functions. Non-convex problems are often more difficult to solve. On this proposed framework, these distributed optimization problems can be solved, thus achieving convergence.Another proposed methods for ADMM are presented in [15] and [16]. In both of these articles is presented two extension of the ADMM, which solve nonconvex and nonsmooth optimization, achieving global convergence of the state.

In general, convex distributed optimization problems have been extensively studied and algorithms are currently have been developed, [17],[18],[19],[20],[21],[22],[23], [24],[25],[26],[27],

[28],[29],[30],[31]. While for nonconvex problems, there is a significant lack of solutions. Due to their structure, it is often difficult to find solutions. Fortunately, various methods have been developed in order to solve them. First, a proximate approach based on the primal-dual method is presented in [4]. The proposed approach is based on the primal-dual method, where the primal step minimizes a certain approximation of the augmented Lagrangian of the problem, and the dual step performs an approximate dual ascent, [32]. In addition, this proposed method transforms the nonconvex function into a strongly convex, thus creating a more linearized approach into solving the problem. The primal dual method is also used in [33]. Another primal-dual approach has been researched in [34]. It is similar to [4], but with different constraints and algorithm analysis. In [34] is proposed an Accelerated Distributed Augmented Lagrangian method which has a faster convergent rate than a simple Augmented Lagrangian method. Another approach to solving nonconvex problems was studied by Paolo Di Lorenzo and Gesualdo Scutari in [35],[36],[37]. A new method has been developed, named NEXT, which was the first algorithmic framework for distributed optimization of a sum of nonconvex (possibly and nonsmooth) cost functions. The described method uses a continuous convex approximation technique, which uses dynamic consensus to distribute computation among agents. Each agent first solves a local convex approximation of the non-convex problem (possibly inexactly) and then performs a local averaging operation. In [38] was studied a general framework for nonconvex distributed optimization. To achieve convergence, a perturbed method was employed in the analysis, thus achieving optimal consensus among the agents. In [39] and [40], two distributed techniques are being presented. In [39] uses nonconvex constraints to bound the velocities of the agents with non-uniform position constrains. Based on a switching method is guaranteed boundedness of the state with sampled neighbor information. The same problem is solved in [40], with the only difference being the topology of the graph, which shifts overtime. Another research in nonconvex optimization is presented in [41]. The objective is to steer the agents into an optimal point using partial information which is related to input and output. In addition, momentum-based optimal coordinators are design in order to achieve a steady-state. When solving an optimization problem, it is common for the solver to reach a local minimum. This problem is presented in [42] with a solution for escaping local optima. The concept is to transform the gradient method into a boosted gradient with nonzero magnitude. Thus, the optimizer will converge into a global minimum. In [43, 44] is presented a nonconvex framework for solving online the optimization problem using dynamic regrets. In this proposed framework, [45], is presented an asynchronous distributed approach in which both objective function and constraints are nonconvex. It uses a trigger event on which every node has limited access to the function and the constraints. During that time, the method of multipliers is applied to minimize the Augmented Lagrangian. Nonetheless, in some nonconvex optimization problems appear also and nonlinearly coupled information as in [46]. A predifined-time approach was deisgned in order to solve the distributed problem, thus achieving a local optimal solution.

Most optimization algorithms use gradients to be solved. However, there exists a number of algorithms who are gradient free. These algorithms from their nature are stochastic. An interesting research was conducted by Hajinezhad and Zavlanos in [47], where the distributed nonconvex optimization is being solved without the use of gradient. Their approach is based on a primal dual method, in which every agent has access only to partial information about the global cost function, knowing only the zeroth-order information of their local cost function.

1.2.3 Graph Topology Variations

While the objective function fulfills a significant role in distributed optimization of the agents, there is another factor that contributes towards the problem. It is also important to consider the topology of the graph. Most of the above mention researches have an undirected topology considering the graph. In [22], [31], [48], [49], [50] and [51] the topology of the graph is directed. The main categorization of the topology lies into static and dynamic graphs. The static graphs have one topology that does not change in time. While dynamic topology (or switching topology) changes in time. Therefore, the matrices and vectors of the problem are time variant. In fact, it's clear from [52] that different ways of solving the optimization problem exist, regarding the topology. Throughout the graph is observed the communication tradeoff of the agents. Thus, network topology fulfills a crucial role in the analysis.

Although fixed topologies have been extensively studied throughout the years, research has been conducted on switching topologies as well. A thoroughly established illustration of the usage of both fixed and switching topology is studied in [53]. A class of distributed non-linear controllers have been designed in order to achieve consensus for high-order agents in both topologies. In [54] a similar approach is presented, but with unknown high frequency gain signs. The method which was proposed assures consensus between the agents with non-identical unknown control directions under the switching graph. A more probabilistic approach is presented in [55]. While exchanging packages, the agents change their topology randomly when a packet is lost. As a result, the agents are modelled as a system with a stochastic switching undirected topology, which in conjunction with the proposed algorithm is robust to ensure consensus. Oftentimes, consensus protocols tend to be executed asynchronous between the agents. Such cases are presented in [56] and [57]. In the first one, a switched observer was designed which estimates the state information of the leader, which is based on a general switching topology rule of the spanning tree of the graph. While feedback distributed control laws ensure consensus in all modes of the graph. The later one, ensures consensus between the agents, while the topology switches during time-delays. In [58] is presented a novel approach regarding the losses which occurs during the switching of the topology. This simple approach is base on a high-order filter to generate differentiable control variables using neighbor position information and discontinuous topology information. Another research paper is presented in [59], which ensures robust consensus tracking

control, considering the fact that the topology changes in time with disturbances in the communications. Lastly, in [60] is presented an adaptive mechanism which ensures cooperation between the agents under switched topologies. In this paper are presented two distributed event-trigger control schemes, in which the agents track reference inputs asymptotically and reject disturbances.

1.3 Thesis Structure

This diploma thesis consists of six chapters in total. The first chapter contains the introduction for this thesis and the motivation behind it. In addition, the chapter also contains the review of the literature. The chapter also contains the thesis structure and bibliography.

The second chapter discusses the fundamental mathematical concepts and axioms utilized for proving and designing both the algorithm and the robust distributed control laws.

In the third chapter, the NOCGPROX variables are introduced in order to solve the proximal output consensus. Additionally, it contains a mathematical proof analysis of the algorithm with its lemmas and theorem, which prove proximal consensus.

In the fourth chapter, the nonlinear dynamics of the agents are transformed to incorporate the NOCGPROX variables into the dynamics. The robust distributed control laws are introduced which ensure the square integration of the variables with sliding surface feedback linearization method to ensure that the variables tends to zero and be square integrable.

The fifth chapter is consisted of the simulation of the algorithm and control laws for the agents. In the simulation, there are applied different scenarios in order to achieve optimal consensus of the agents.

The sixth and final chapter contains the concluding remarks of this thesis and purpose, while future research is discussed for improvements of this work.

Chapter 2

Mathematical Background

In this chapter, the mathematical background of this thesis is presented. Specifically, is introduced the basic notations and definitions of graph theory, as well some mathematical concepts of matrix theory, Lyapunov stability theory and basic theorems of mathematical analysis. In addition, is introduced and the augmented Lagrangian algorithm.

2.1 Graph Theory

2.1.1 Basic Definitions

Definition 2.1. [61] A finite graph $\mathcal{G}$ is a non-empty finite set $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ which is consisted of a finite non-empty set $\mathcal{V}(\mathcal{G})$, a finite non-empty set $\mathcal{E}(\mathcal{G})$ and $\mathcal{A}$ is the incidence matrix of the graph. The set $\mathcal{V}(\mathcal{G})$ contains the vertices (nodes) of the graph, with $|\mathcal{V}| = N$ and the set $\mathcal{E}(\mathcal{G})$ contains the edges of the graph with $|\mathcal{E}| = E$, which connect the vertices.

If there exists an edge between the vertices v_i and v_j, then the vertices are defined as adjacent with the symbol $v_i \sim v_j$.

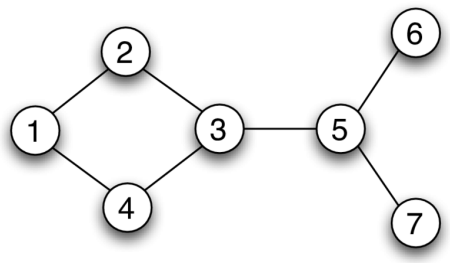

Figure 2.1: Undirected graph topology.

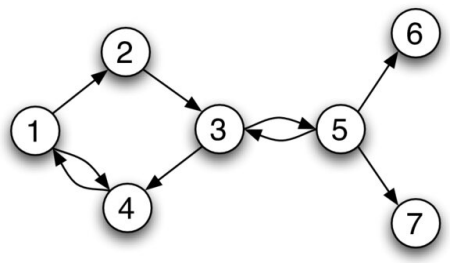

Figure 2.2: Directed graph topology.

Definition 2.2. [61] A neighborhood of an $i-$agent is defined as
$\mathcal{N}(i) := \{j \in \mathcal{V} \,|\, (i,j) \in \mathcal{E}\} \subset \mathcal{V}$.

Definition 2.3. [61] A path of length l from $\{v_{i_0}, v_{i_1}, ..., v_{i_l}\}$ in a graph is a sequence of $l+1$ distinct nodes starting with v_{i_0} and ending with v_{i_l} such that consecutive nodes are adjacent.

Definition 2.4. [61] A cycle is a connected graph where every node has exactly two neighbors.

Definition 2.5. [61] A graph $\mathcal{G}$ is connected, if for every pair of nodes of $\mathcal{V}(G)$, exists a path which have this pair as an end.

Definition 2.6. [61] If in a graph, in addition to its nodes and edges, exists a function $\mathcal{W} : E \to \mathbf{R}$ where assigns a value in every edge, then the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$ is called weighted graph.

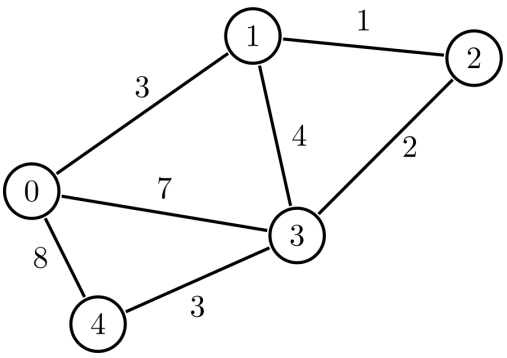

Figure 2.3: Undirected weighted graph topology.

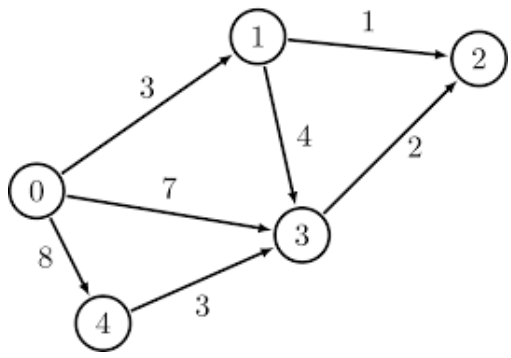

Figure 2.4: Directed weighted graph topology.

2.1.2 Graph Matrices

In addition to the basic graph notations, matrices can describe them in an algebraic form. Due to the selected algorithm, an incidence matrix is used.

Definition 2.7. [61] The degree $d(v_i)$ of a node in an undirected graph $\mathcal{G}$ is the total number of nodes which are adjacent to v_i.

Definition 2.8. [61] The degree matrix D of an undirected graph $\mathcal{G}$ is a diagonal matrix whose elements are the degree nodes $d(v_i)$.

Definition 2.9. [32] The incidence matrix $\mathcal{A}$ is a $N \times E$ matrix, which describes the relationship between a node and an edge. In its $i-$th row and $j-$th column has the number of times the $j-$th edge is incident with the $i-$th vertex. Now the matrix is defined as

$$[\mathcal{A}]_{ev} = \begin{cases} 1, & if \;\; (i,j) \in \mathcal{E}, with \;\; i > j \;\; and \;\; v = i \\ -1, & if \;\; (i,j) \in \mathcal{E}, with \;\; i > j \;\; and \;\; v = j \\ 0, & otherwise \end{cases} \tag{2.1.1}$$

Definition 2.10. [61] The Laplacian matrix of an undirected graph $\mathcal{G}$ is defined by

$$L = D - \mathcal{A} \tag{2.1.2}$$

where D is the degree matrix and $\mathcal{A}$ is the incidence matrix.

Definition 2.11. [32] The signed Laplacian matrix of a graph $\mathcal{G}$ is defined as

$$L_- = \mathcal{A}^T\mathcal{A} \tag{2.1.3}$$

Definition 2.12. [32] The signless Laplacian matrix of a graph $\mathcal{G}$ is defined as

$$L_+ = 2D - \mathcal{A}^T\mathcal{A} \tag{2.1.4}$$

2.2 Matrix Theory

In this section are going to be introduced the basic notations of matrix theory that will be used.

Definition 2.13. [62] Consider a vector field of $\mathbf{R}^n$ and a squared matrix $A \in \mathbf{R}^{n\times n}$. Then, for a vector $x \in \mathbf{R}^n$ and a constant $\lambda \in \mathcal{C}$ holds that

$$Ax = \lambda x \tag{2.2.1}$$

where x is called eigenvector of matrix A which is related to the eigenvalue λ.

The matrix A can be inverted, if he does not have a zero eigenvector. The eigenvalues of a matrix are computed from

$$\chi_A(\lambda) = det(\lambda I_n - A) = 0 \tag{2.2.2}$$

This equation is called the characteristic polynomial of the matrix A, with the eigenvalues being its roots. The algebraic multiplicity of an eigenvalue λ is the multiplicity as a root of the equation (2.2.2). The geometric multiplicity of an eigenvalue λ, is the maximum number of linear independent eigenvectors, which are associated with this eigenvalue.

Theorem 2.1. [62] Let S be a matrix of eigenvectors of a given square matrix A and Λ be a diagonal matrix with the corresponding eigenvalues on the diagonal. Then, as long as S is a square matrix, matrix A can be written as an eigen decomposition

$$A = S\Lambda S^{-1} \tag{2.2.3}$$

where Λ is a diagonal matrix. Furthermore, if A is symmetric, then the columns of S are orthogonal vectors.

Definition 2.14. [62] A matrix A is called semipositive if the quadratic form

$$x^T Ax \geq 0, \forall x$$

and positive if the quadratic form is

$$x^T Ax > 0, \forall \ x \neq 0$$

Corollary 2.1. [62] A matrix A is semipositive iff all his eigenvalues are nonnegative. A matrix A is positive if all his eigenvalues are positive.

Definition 2.15. [62] A matrix $A \in \mathbf{R}^{n\times n}$ is defined as nonnegative if his elements are nonnegative, i.e. $\alpha_{ij} \geq 0$, for $i, j = 1, ..., n$.

Definition 2.16. [62] The Kronecker product of two matrices $A \in \mathbf{R}^{n\times m}$ and $B \in \mathbf{R}^{p\times q}$, with $\alpha_{ij} = [A]_{ij}$ and $b_{ij} = [B]_{ij}$, is written as $A \otimes B$ and defined as an $np \times mq$ matrix below

$$A \otimes B = \begin{bmatrix} \alpha_{11}B & \cdots & \alpha_{1m}B \\ \alpha_{21}B & \cdots & \alpha_{2m}B \\ \vdots & \vdots & \vdots \\ \alpha_{n1}B & \cdots & \alpha_{nm}B \end{bmatrix} \tag{2.2.4}$$

The properties of the Kronecker product are presented below

$$A \otimes (B + C) = A \otimes B + A \otimes C$$

$$(kA) \otimes B = A \otimes (kB) = k(A \otimes B)$$

$$(A \otimes B) \otimes C = A \otimes (B \otimes C)$$

$$A \otimes 0 = 0 \otimes A = 0$$

where matrices A, B, C have the appropriate dimensions, 0 is the zero matrix and k is constant.

2.3 Mathematical Analysis

In this section are presented the basic notations, theorems and definitions of mathematical analysis. Specifically of real analysis, functional analysis and convex and non-Convex analysis.

Definition 2.17. [63] A continuous function $a : \mathbf{R}_+ \to [0, \infty)$ is a class $\mathcal{K}$ function if

- $a \in C^0(\mathbf{R}_+)$

- it is strictly increasing in $\mathbf{R}_+$
- $a(0) = 0$

Definition 2.18. [63] A continuous function $a : \mathbf{R}_+ \to [0, \infty)$ is a class $\mathcal{K}_\infty$ function if

- $a \in C^0(\mathbf{R}_+)$
- it is strictly increasing in $\mathbf{R}_+$
- $a(0) = 0$
- $\lim_{s\to\infty} a(s) = \infty$

Definition 2.19. [63] A continuous function $a : \mathbf{R}_+ \times \mathbf{R}_+ \to \mathbf{R}_+$ is a class $\mathcal{KL}$ function if

- $a(\cdot, t)$ is strictly increasing in $\mathbf{R}_+$, $\forall t \geq 0$
- $a(s, \cdot)$ is strictly decreasing in $\mathbf{R}_+$, $\forall s \leq 0$
- $\lim_{t\to\infty} a(s, t) = 0, \forall s \geq 0$
- $a(0, t) = 0, \forall t \geq 0$

Definition 2.20. [64] A continuous function $W : \mathbf{R}^n \to \mathbf{R}$ is positive defined iff

- $W(0) = 0$
- $W(x) > 0, \quad \forall x \in \mathbf{R}^n - \{0\}$
- exists $r > 0$, such that $\inf_{||x||\geq r} W(x) > 0$

Lemma 2.1. [64] (Young inequality) For all $a, b \in \mathbf{R}$ and $\epsilon > 0$ it holds that

$$a \cdot b \leq \frac{\epsilon}{2}a^2 + \frac{1}{2\epsilon}b^2 \tag{2.3.1}$$

Lemma 2.2. [63] (Hölder's inequality) Assume that p and q are in the open interval $(1, \infty)$ with $1/p + 1/q = 1$. Then

$$\left(\sum_{i=1}^{n} |a_i b_i|\right) \leq \left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |b_i|^q\right)^{1/q} \tag{2.3.2}$$

For the case $|a_i| = 1$ for all $i = 1, ..., n$ we have

$$\left(\sum_{i=1}^{n} |b_i|\right) \leq n^{1-p} \left(\sum_{i=1}^{n} |b_i|^p\right)$$

Definition 2.21. [65] A function $f \in C^1(\mathbf{R}^n)$ is called $L-$smooth if satisfies the below inequality, $\forall x_1, x_2 \in \mathbf{R}^n$:

$$||\nabla f(x_1) - \nabla f(x_2)|| \leq L||x_1 - x_2|| \tag{2.3.3}$$

or

$$|f(x_1) - f(x_2) - \nabla f(x_2)^T(x_1 - x_2)| \leq \frac{L}{2}||x_1 - x_2||^2 \tag{2.3.4}$$

Definition 2.22. [65] A function $f \in C^1(\mathbf{R}^n)$ is called convex, if $\forall x_1, x_2 \in \mathbf{R}^n$:

$$f(x_1) - f(x_2) \geq f(x_2)^T(x_1 - x_2) \tag{2.3.5}$$

Definition 2.23. [65] A function $f \in C^1(\mathbf{R}^n)$ is called $\sigma-$strongly convex, with $\sigma > 0$, if $\forall x_1, x_2 \in \mathbf{R}^n$:

$$f(x_1) - f(x_2) - f(x_2)^T(x_1 - x_2) \geq \frac{\sigma}{2}||x_1 - x_2||^2 \tag{2.3.6}$$

if $f \in C^2(\mathbf{R}^n)$, is strongly convex and smooth then

$$\sigma I_n \preceq \nabla^2 f(x) \preceq L I_n \tag{2.3.7}$$

where $I_n \in \mathbf{R}^{n\times n}$ is the unit matrix with size n. If $f(x_1) - f(x_2) - f(x_2)^T(x_1 - x_2) \geq 0$, then from (2.3.4) and (2.3.6) is obtained:

$$\frac{\sigma}{2}||x_1 - x_2||^2 \leq f(x_1) - f(x_2) - f(x_2)^T(x_1 - x_2) \leq \frac{L}{2}||x_1 - x_2||^2 \tag{2.3.8}$$

Definition 2.24. [65] A function $f \in C^1(\mathbf{R}^n)$ is called Lipschitz, if $\forall x_1, x_2 \in \mathbf{R}^n$ and a constant $\mathbf{L} \geq 0$:

$$||f(x_1) - f(x_2)|| \leq \mathbf{L}||x_1 - x_2|| \tag{2.3.9}$$

Definition 2.25. [65] A function $f(x)$ is called locally Lipschitz in $\mathcal{D} \subset \mathbf{R}^n$, if every point of $\mathbf{D}$ has an area $\mathbf{D}_0$ which satisfies the Lipschitz condition for a Lipschitz constant $\mathbf{L}_0$.

Definition 2.26. [65] A function $f(x)$ is called globally Lipschitz if it is Lipschitz on $\mathbf{R}^n$.

Theorem 2.2. [66] If $f : R^n \to R$ is a Lipschitz continuously differentiable on a convex set $\mathcal{C}$ $(f \in C^1(\mathcal{C}))$ with some Lipschitz constant $\mathbf{L}$, then $\phi(x) = f(x) - \frac{1}{2}ax^Tx$ is a convex function on $\mathcal{C}$ for every $a \leq -\mathbf{L}$.

2.4 Lyapunov Stability Theory

Lyapunov theory was invented from the Russian mathematician Aleksandr Lyapunov, in order to study the stability of dynamical systems. Through his findings, he was able to conclude if a dynamical system is either stable or unstable. The basic concept of stability theory is the definition of a suitable scalar function, which is analyzed to evaluate its properties.

Consider the $n-$dimensional system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) \tag{2.4.1}$$

where $f : \mathcal{D} \to \mathbf{R}^n$ is locally Lipschitz over a domain $\mathcal{D} \subset \mathbf{R}^n$. Suppose $\boldsymbol{x}^*$ is an equilibrium point of (2.4.1); that is $f(\boldsymbol{x}^*) = 0$.

Definition 2.27. [64] Let $f(x)$ be a locally Lipschitz defined over a domain $\mathcal{D} \subset \mathbf{R}^n$, which contains the origin, and $f(0) = 0$. The equilibrium point $x^* = 0$ 0f (2.4.1) is

- stable if for each $\varepsilon > 0$ there is $\delta > 0$ (dependent on ε) such that

$$||x(0)|| < \delta \Rightarrow ||x(t)|| < \varepsilon, \forall t \geq 0$$

- unstable if it is not stable.

- asymptotically stable if it is stable and δ can be chosen such that

$$||x(0)|| < \delta \Rightarrow \lim_{t\to\infty} x(t) = 0$$

- exponentially stable if there exists δ, M, σ such that

$$||x(0)|| < \delta \Rightarrow ||x(t)|| \leq Me^{-\sigma t}||x(0)||, \ \forall t \geq 0$$

Definition 2.28. [64] Consider a function $V : \mathbf{R}^n \to R$, which:

- $V(0) = 0$

- $V(x) > 0$, $\forall x \neq 0$

then it is a positive define function. This function is called Candidate Lyapunov function.

Definition 2.29. [64] A Lyapunov function $V : \mathbf{R}^n \to R$ is called radially unbounded, if $\forall M > 0$ the set $\{x \in \mathbf{R}^n : \ V(x) \leq M\}$ is bounded, or

$$\lim_{||x||\to+\infty} V(x) = +\infty$$

Theorem 2.3. [64] Consider a positive defined function $V \in C^1(\mathbf{R}^n)$. Consider an open set $\mathcal{D} \subset \mathbf{R}^n$, with the 0 inside it such that

$$\nabla V(x) f(x) \leq 0, \forall x \in \mathcal{D}$$

then, $0 \in \mathbf{R}^n$ is stable for the system of (2.4.1).

Theorem 2.4. [64] Consider a positive defined function $V \in C^1(\mathbf{R}^n)$. Consider an open set $\mathcal{D} \subset \mathbf{R}^n$, with the 0 inside it such that

$$\nabla V(x) f(x) < 0, \forall x \in \mathcal{D}, \forall x \neq 0$$

then, $0 \in \mathbf{R}^n$ is asymptotically stable for the system of (2.4.1).

Definition 2.30. [64] The equilibrium point of (2.4.1) is

- uniformly bounded if there exists a positive constant c, independent of $t_0 \geq 0$, and for every $a \in (0, c)$, there exists $\beta = \beta(a) > 0$, independent of t_0, such that

$$||x(t_0)|| \leq 0 \Rightarrow ||x(t)|| \leq \beta, \forall t \geq t_0 \tag{2.4.2}$$

- globally uniformly bounded if (2.4.2) holds for arbitrarily large a.
- uniformly ultimately bounded with ultimate bound b if there exists positive constants b and c, independent of $t_0 \geq 0$, and for every $a \in (0, c)$, there is $T = T(a, b) \geq 0$, independent of t_0, such that

$$||x(t_0)|| \leq a \Rightarrow ||x(t)|| \leq b, \forall t \geq t_0 + T \tag{2.4.3}$$

- globally uniformly ultimately bounded if (2.4.3) holds for arbitrarily large a.

Lemma 2.3. [64] Let $V : \mathcal{D} \to \mathbf{R}$ be a continuous positive function defined inside the domain $\mathcal{D} \subset \mathbf{R}^n$ that contains the origin. Let $B_r \subset \mathcal{D}$ for some $r > 0$. Then, there exist class $\mathcal{K}$ functions a_1 and a_2, defined on $[0, r]$, such that

$$a_1(||x||) \leq V(x) \leq a_2(||x||) \tag{2.4.4}$$

$\forall x \in B_r$. If $\mathcal{D} = \mathbf{R}^n$, the functions a_1 and a_2 will be defined on $[0, \infty)$ and the foregoing inequality will hold $\forall x \in \mathbf{R}^n$. Moreover, if $V(x)$ is radially unbounded, then a_1 and a_2 can be chosen to belong to class $\mathcal{K}_\infty$.

2.5 Prox-GPDA

Consider a network of N agents and the following optimization problem

$$\min_{x \in \mathbf{R}^n} g(x) = \sum_{i=1}^{N} f_i(x) \tag{2.5.1}$$

where $f_i(x) : \mathbf{R}^n \to \mathbf{R}$, $i \in \{1, ..., N\} := [N]$ are the nonconvex local cost functions of each agent, which are assumed to be smooth and have Lipschitz continuous gradients. The agents

exchange information only with their neighbors (see definition 2.2) in a distributed manner with $|\mathcal{N}_i| = d_i$. The communication graph is assumed to be undirected and connected (see definition 2.5). The distributed version of problem (2.5.1) is

$$\min_{x_i \in \mathbf{R}^n} f(x) := \sum_{i=1}^{N} f_i(x_i), \ s.t. \ x_i = x_j, \ \forall (i,j) \in \mathcal{E} \tag{2.5.2}$$

Augmented Lagrangian Algorithms have a primal-dual multiplier form, [67], and are typically used to incorporate the constraints of the problem. The algorithm which will be used in this dissertation is Prox-GPDA, [4, 32], which is a proximal form of the classic augmented Lagrangian algorithm with an added proximal term.

Obviously, the problem (2.5.2) is equivalent to the problem (2.5.1) as long as the graph $\mathcal{G}$ is connected. For notational simplicity, the sequence $x := \{x_i\} \in \mathbf{R}^{Nn \times 1}$, and $Q = N \times n$ are defined. The incidence matrix of the communication graph $\mathcal{A}$ is introduced in Definition 2.9. In order for the dimensions to be matched, we define the extended incidence matrix as

$$A = \mathcal{A} \otimes I_n \in \mathbf{R}^{En \times Q} \tag{2.5.3}$$

In the same manner, the degree matrix is defined as

$$\mathbf{D} = D \otimes I_n \in \mathbf{R}^{Q \times Q} \tag{2.5.4}$$

Lastly, the Laplacian matrices are defined in Definitions 2.11 and 2.12. The extended form of the signed and the signless Laplacian are:

$$L_- = L_- \otimes I_n \in \mathbf{R}^{Q \times Q}, L_+ = L_+ \otimes I_n \in \mathbf{R}^{Q \times Q} \tag{2.5.5}$$

Using the notations above, the problem 2.5.2 can be rewritten in the following compact form

$$\min_{x \in \mathbf{R}^Q} f(x), \ \ s.t. \ Ax = 0 \tag{2.5.6}$$

Prox-GPDA is built upon the classical Augmented Lagrangian Method [67]. The augmented Lagrangian function is defined as

$$L_\beta(x, \mu) = f(x) + \langle \mu, Ax \rangle + \frac{\beta}{2} ||Ax||^2 \tag{2.5.7}$$

where $\mu \in \mathbf{R}^Q$ is the Lagrangian dual variable; $\beta > 0$ is a penalty parameter. Let the matrix $B \in \mathbf{R}^{Q \times Q}$ which is some arbitrary matrix that will be defined in the analysis. The algorithm Prox-GPDA introduced in [32] is shown below.

Algorithm 1 The Proximal Gradient Primal Dual Algorithm (Prox-GPDA)

Input: $\mu^0 = 0$, $x^0 \in \mathbf{R}^Q$
Output: $x^{r+1} \in \mathbf{R}^Q$

At each iteration $r+1$, update variables by:

$$x^{r+1} = arg \min_{x \in \mathbf{R}^Q} \langle \nabla f(x^r), x - x^r \rangle + \langle \mu^r, Ax \rangle + \frac{\beta}{2}||Ax||^2 + \frac{\beta}{2}||x - x^r||^2_{B^T B} \tag{2.5.8a}$$

$$\mu^{r+1} = \mu^r + \beta A x^{r+1} \tag{2.5.8b}$$

As seen in algorithm 1, the cost function has been replaced with a linear gradient step, in order to solve $x-$subproblem (2.5.8a). This is efficient, hence does not require computations of the zeroth order of the function $f(x)$. It is also advantageous in cases where the cost function is not available. Furthermore, the proximal term which is introduced in the algorithm is $\frac{\beta}{2}||x - x^r||^2_{B^T B}$. This term is crucial in both the algorithm implementation and analysis, since it ensures that:

1. The primal problem transforms from nonconvex to strongly convex.

2. The primal problem is decomposable over different network nodes, thus distributedly implementable.

In order to ensure the strong convexity property, the matrix $B^T B$ must be chosen such that $A^T A + B^T B \succeq I$. Then, for any $\beta > 0$ the objective function of the $x-$subproblem is strongly convex. Moreover, a globally Lipschitz gradient $f(x)$ is necessary for the convergence of the algorithm.

Assumption 1. The function $f(x)$ is differentiable and has Lipschitz continuous gradient, i.e.

$$||\nabla f(x) - \nabla f(y)|| \leq \mathrm{L}||x - y||, \ \ \forall x, y \in \mathbf{R}^Q$$

Further, it is assumed that $A^T A + B^T B \succeq I_Q$.

Assumption 2. There exists a constant $\delta > 0$ such that

$$\exists \ \underline{f} > -\infty, \ \ s.t. \ \ f(x) + \frac{\delta}{2}||Ax||^2 \geq \underline{f}, \ \ \forall x \in \mathbf{R}^Q$$

Without loss of generality and for the simplicity of notations, is going to be assumed that $\underline{f} = 0$.

Assumption 3. The constraint $Ax = 0$ is feasible over $x \in \mathbf{R}^Q$.

Assumption 4. The network topology is described by an undirected connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which is composed of N agents (nodes) and is connected by E edges.

With these basic assumptions for the algorithm, B is chosen such that $B := |A|$, or *signless incidence matrix*. Thus, it is known that $L_- = A^T A$, then $L_+ = B^T B$. Then it is certain that $A^T A + B^T B = 2\mathrm{D}$. So the minimum singular value of the matrix $A^T A$ is positive and is depended by the degrees of the graph. From equation (2.5.8a), the optimality condition is derived as

$$\nabla f(x^r) + A^T \mu^r + \beta A^T A x^{r+1} + \beta B^T B(x^{r+1} - x^r) = 0 \tag{2.5.9}$$

Rearranging the equation (2.5.9), the proximal update of the primal variable is

$$\nabla f(x^r) + A^T \mu^r + \beta A^T A x^{r+1} + \beta B^T B x^{r+1} - \beta B^T B x^r = 0$$

$$\Rightarrow \nabla f(x^r) + A^T \mu^r + \beta \left(A^T A + B^T B\right) x^{r+1} - \beta B^T B x^r = 0$$

$$\Rightarrow \nabla f(x^r) + A^T \mu^r + 2\beta \mathrm{D} x^{r+1} - \beta L_+ x^r = 0$$

$$\Rightarrow 2\beta \mathrm{D} x^{r+1} = \beta L_+ x^r - A^T \mu^r - \nabla f(x^r)$$

Therefore, in vector form the update of the primal variable is

$$x^{r+1} = \frac{1}{2\beta} \mathrm{D}^{-1} \left[\beta L_+ x^r - A^T \mu^r - \nabla f(x^r)\right] \tag{2.5.10}$$

or every agent is computing the position with

$$x_i^{r+1} = \frac{1}{2} x_i^r + \frac{1}{2\beta d_i} \left[\beta \sum_{j \in \mathcal{N}(i)} x_j^r - \nabla f_i(x_i^r) - \beta \sum_{j \in \mathcal{N}(i)} \sum_{\ell=1}^{r-1} \left(x_i^\ell - x_j^\ell\right)\right] \tag{2.5.11}$$

Thus, the primal update is computed by equation (2.5.11).

Chapter 3

Optimal Consensus via NOCGPROX Variables

In this chapter, the NOCGPROX variables are introduced which will serve as suitable error variables for the agent outputs. It is rigorously proven that the regulation of these variables guarantees optimal consensus for all agent outputs.

3.1 Optimal Consensus Problem

Consider a group of N agents with output vectors $x_i \in \mathbf{R}^M$ and local cost functions $f_i : \mathbf{R}^M \to \mathbf{R}$ as in [7]. We aim to drive all agent outputs to $x^* \in \mathbf{R}^Q$ which is the optimal solution of (2.5.6). Apart from the assumptions of algorithm 1, the following assumption is also considered.

Assumption 5. The agents communicate with each other only at sampled times, i.e. every time instant $t = kT$, $k \in N$ where T is the sampled period. In that time period, the agent i will get samples of the output $x_j(kT)$ from his neighbors $j \in \mathcal{N}(i)$. In order to allow for communication delays during the exchange of the information, it is assumed that every measurement $x_j(kT)$ from the j agents is obtainable by i agent in time, $t_e = kT + d_{max}$, with $d_{max} \geq 0$ be the maximum permitted communication delay. Furthermore, it is assumed that the sampling period is selected sufficiently large, namely $T > d_{max}$.

3.1.1 Definition of NOCGPROX Variables

In order to solve the optimal consensus problem, we define auxiliary variables that recast the described above optimization problem into a regulation one. Since every agent exchanges its output measurements in discrete time intervals, the variables can utilize only samples

from neibhoring agents. The proposed regulation variables are called NOn-Convex Gradient PROXimal (NOCGPROX) variables and are defined as

$$q_i(t) := x_i(t) - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) - \left[1 - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right)\right] s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right) \tag{3.1.1}$$

where the discrete-time vector s_i is defined as:

$$s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) := \frac{1}{2} x_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) + \frac{1}{2\beta d_i}\left[\beta \sum_{j\in\mathcal{N}(i)} x_j\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) - \nabla f_i\left(x_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right)\right)\right.$$
$$\left. - \beta \sum_{j\in\mathcal{N}(i)} \sum_{\ell=1}^{\lfloor t/T \rfloor} \left(x_i(\ell T) - x_j(\ell T)\right)\right] \tag{3.1.2}$$

for every $i \in \mathcal{V}$ with $\beta > 0$. We will prove that if the variables converge to zero and are also integrable (i.e. $\mathcal{L}_1$), then the agents achieve consensus and converge to the optimal point.

The function $\rho : [0, +\infty) \to [0, 1]$ is a continuous function that is employed to ensure the continuity of q_i by defining a continuous convex combination of the two recent samples $s_i(kT), s_i((k-1)T)$ [6]. A suitable function is presented in [10]:

$$\rho(t) := \begin{cases} 0, & t \in [0, d_{max}) \\ \frac{t-d_{max}}{T-d_{max}} - \frac{1}{2\pi}\sin\left(2\pi \frac{t-d_{max}}{T-d_{max}}\right), & t \in [d_{max}, T] \\ 1, & t \in (T, +\infty) \end{cases} \tag{3.1.3}$$

By definition, the function ρ is increasing in its domain and is continuously differentiable for $t \geq 0$. Both derivates $\frac{d\rho}{dt}$ and $\frac{d^2\rho}{dt^2}$ are also continuous in their domain. In the time interval $[0, d_{max}]$ the auxiliary function is equal to zero. As a result, the NOCGPROX variables will only use neighbors' samples after d_{max} time from the last sampling time.

Remark 3.1. The function $\rho(t - \lfloor t/T \rfloor T)$ is not continuous for $t = kT$, $k \in \mathbf{N}$. However, if $x_i(t)$ is two times continuous differentiable then one can prove that the variables $q_i(t)$ are also two times continuous differentiable. This is obvious for $t \neq kT$. For $t = kT$ we have

$$\lim_{t\to kT^-} q_i(t) = x_i(kT) - [1 - \rho(T)]s_i((k-2)T) - \rho(T)s_i((k-1)T) = x_i(kT) - s_i((k-1)T)$$
$$\lim_{t\to kT^+} q_i(t) = x_i(kT) - [1 - \rho(0)]s_i((k-1)T) - \rho(0)s_i(kT) = x_i(kT) - s_i((k-1)T)$$

Furthermore, since $d^r\rho(t)/dt^r|_{t=kT} = 0$, for $r = 1, 2$, the variables q_i have also continuous derivates with $q_i^{(r)}(kT) = x_i^{(r)}(kT)$.

Remark 3.2. For $t \in (kT, (k+1)T)$, the variables have value $q_i(t) = x_i(t) - [1 - \rho(t - kT)]s_i((k-1)T) - \rho(t - kT)s_i(kT)$. From definition of $\rho(t)$ in (3.1.3), $\rho(t - kT) = 0, \forall t \in$

$(kT, kT + d_{max})$. As a result, the measurement $s_i(kT)$ is needed only at $t \geq kT + d_{max}$ for the computation of $q_i(t)$. Hence, a control law that aims to regulate $q_i(t)$, will require the output measurements of the neighbors $x_j(kT), j \in \mathcal{N}(i)$, after time $kT + d_{max}$. As a result, with the current setup, communication time delays are allowed.

From the definition of the variables and the remark 3.1, it holds that

$$x_i^{r+1} = q_i^{r+1} + s_i^r \tag{3.1.4}$$

From equation (3.1.4) the discrete time system is

$$\begin{aligned} x_i^{\nu+1} = q_i^{\nu+1} - \frac{1}{2}x_i^{\nu} - \frac{1}{2\beta d_i}\Bigg[\beta \sum_{j\in\mathcal{N}(i)} x_j^{\nu} - \nabla f_i\left(x_i^{\nu}\right) \\ - \beta \sum_{j\in\mathcal{N}(i)} \sum_{\ell=1}^{\nu} \left(x_i^{\ell} - x_j^{\ell}\right)\Bigg], i = \{1, ..., N\} \end{aligned} \tag{3.1.5}$$

where $q := col(q_1, q_2, ..., q_N) \in \mathbf{R}^Q$, $\nabla f(x) := col(\nabla f_1(x_1), \nabla f_2(x_2), ..., f_N(x_N)) \in \mathbf{R}^Q$ and $x := col(x_1, x_2, ..., x_N) \in \mathbf{R}^Q$. If we now define:

$$\mu^{\nu} := \beta \sum_{\ell=1}^{\nu} Ax^{\ell} \tag{3.1.6}$$

then

$$\mu^{\nu+1} = \mu^{\nu} + \beta Ax^{\nu+1} \tag{3.1.7}$$

where $\nu \in \mathbf{N}$ and $\mu^0 = 0$. Therefore, the discrete vector form of equation (3.1.5) will be

$$x^{\nu+1} = q^{\nu+1} - \frac{1}{2\beta}\mathrm{D}^{-1}\left[\beta L_+ x^{\nu} - \nabla f(x^{\nu}) - A^T\mu^{\nu}\right] \tag{3.1.8}$$

The discrete-time update can be written as

$$\nabla f(x^{\nu}) + A^T\mu^{\nu} + \beta A^T Ax^{\nu+1} + \beta B^T B(x^{\nu+1} - x^{\nu}) - 2\beta \mathrm{D}q^{\nu+1} = 0 \tag{3.1.9}$$

Eq. (3.1.9) is subsequently used to prove necessary lemmas and theorems for the convergence to the optimal point.

Remark 3.3. From (3.1.9) one directly obtains that

$$x^{\nu+1} - q^{\nu+1} = \arg\min_{x\in\mathbf{R}^Q} \left\{\nabla f(x^{\nu})(x - x^{\nu}) + \langle \mu^{\nu}, Ax\rangle + \frac{\beta}{2}\|Ax\|^2 + \frac{\beta}{2}\|x - x^{\nu}\|^2_{B^TB}\right\}$$

3.2 Convergence Analysis

We will now prove that if the sequence q^ν is ℓ_2 then the discrete-time update shown above will converge to the optimal consensus point. The proof follows the steps of the analysis of Prox-PDA [4] as detailed in [32]. A Lyapunov-like function is used to prove the desired convergence result.

Initially, a bound is obtained for the size of the constraint violation. Let $\underline{\sigma}(A)$ be the minimum eigenvalue of the matrix $A^T A$. Then we have the following result.

Lemma 3.1. Suppose that the assumptions A.1 and A.3 are satisfied. Then, the following bound for the dual variable is true for the proposed μ update.

$$\begin{aligned}\frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^{\nu}\right\|^2 \leq \frac{3\mathrm{L}^2}{\beta\underline{\sigma}^2(A)}\left\|x^{\nu}-x^{\nu-1}\right\|^2 + \frac{12\beta}{\underline{\sigma}^2(A)}\left\|\,\mathrm{D}\left(q^{\nu+1}-q^{\nu}\right)\right\|^2 \\ +\frac{3\beta}{\underline{\sigma}^2(A)}\left\|B^T B\left(\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right)\right\|^2\end{aligned} \tag{3.2.1}$$

Proof. Considering (3.1.7) in (3.1.9) we obtain

$$\nabla f(x^\nu) + A^T\mu^{\nu+1} + \beta B^T B(x^{\nu+1}-x^\nu) - 2\beta \mathrm{D} q^{\nu+1} = 0 \tag{3.2.2}$$

We have

$$\mu^{\nu+1}-\mu^\nu = \beta A x^{\nu+1} \in \Im(A)$$

i.e., the difference between the dual variable iteration lies in the image space of the matrix A. In addition, from the fact that $\mu^0 = 0$, the dual variable itself lies in the column space of the matrix A.

$$\mu^\nu = \beta\sum_{\ell=1}^{\nu} A x^\ell \in \Im(A). \tag{3.2.3}$$

As a result, if we denote by $\underline{\sigma}(A)$ the smallest nonzero singular value of A, then

$$\lambda_{\min}^{1/2}(A^T A)\left\|\mu^{\nu+1}-\mu^\nu\right\| \leq \left\|A^T\left(\mu^{\nu+1}-\mu^\nu\right)\right\|.$$

This inequality, along with equation (3.2.2) and $\lambda_{min}^{1/2}(A^T A) = \underline{\sigma}(A)$ implies that

$$\begin{aligned}\left\|\mu^{\nu+1}-\mu^\nu\right\| &\leq \frac{1}{\underline{\sigma}(A)}\left\|A^T\left(\mu^{\nu+1}-\mu^\nu\right)\right\| \\ &\leq \frac{1}{\underline{\sigma}(A)}\left\|A^T\mu^{\nu+1}-A^T\mu^\nu\right\| \\ &\leq \frac{1}{\underline{\sigma}(A)}\left\|-\beta B^T B\left[\left(x^{\nu+1}-x^\nu\right)-\left(x^\nu-x^{\nu-1}\right)\right]\right. \\ &\quad \left.-\left[\nabla f(x^\nu)-\nabla f(x^{\nu-1})\right]+2\beta\mathrm{D}\left(q^{\nu+1}-q^\nu\right)\right\|\end{aligned}$$

The next step is to square both sides and divide by β. Then it is

$$\frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^{\nu}\right\|^2 \leq \frac{1}{\beta\underline{\sigma}^2(A)}\left\|\beta B^T B\left[\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right]\right.$$
$$\left.+\left[\nabla f(x^{\nu})-\nabla f(x^{\nu-1})\right]-2\beta \mathrm{D}\left(q^{\nu+1}-q^{\nu}\right)\right\|^2$$

By applying the triangle inequality, we have

$$\frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^{\nu}\right\|^2 \leq \frac{1}{\beta\underline{\sigma}^2(A)}\left[\left\|\beta B^T B\left[\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right]\right\|\right.$$
$$\left.+\left\|\nabla f(x^{\nu})-\nabla f(x^{\nu-1})\right\|+\left\|\beta \mathrm{D}\left(q^{\nu+1}-q^{\nu}\right)\right\|\right]^2$$

Now utilizing Hölder's inequality , (Lemma 2.2) and assumption A.1 we obtain

$$\frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^{\nu}\right\|^2 \leq \frac{1}{\beta\underline{\sigma}^2(A)}\left[3\beta^2\left\|B^T B\left[\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right]\right\|^2\right.$$
$$\left.+3\mathrm{L}^2\left\|x^{\nu}-x^{\nu-1}\right\|^2+12\beta^2\left\|\mathrm{D}\left(q^{\nu+1}-q^{\nu}\right)\right\|^2\right]$$

□

The second step of the analysis is the boundedness of the augmented Lagrangian function. Then the later lemma is proposed.

Lemma 3.2. Suppose that the A.1 and A.3 are satisfied. Then the following expression is true for the proposed error variables $q_i(t)$ (3.1.1)-(3.1.2).

$$\begin{aligned} L_\beta(x^{\nu+1},\mu^{\nu+1})-L_\beta(x^{\nu},\mu^{\nu}) \leq & -\frac{\gamma-2\mathrm{L}-\epsilon}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2+\frac{2\beta^2}{\epsilon}\left\|\mathrm{D}q^{\nu+1}\right\|^2 \\ & +\frac{3\mathrm{L}^2}{\beta\underline{\sigma}^2(A)}\left\|x^{\nu}-x^{\nu-1}\right\|^2+\frac{12\beta}{\underline{\sigma}^2(A)}\left\|\mathrm{D}(q^{\nu+1}-q^{\nu})\right\|^2 \\ & +\frac{3\beta}{\underline{\sigma}^2(A)}\left\|B^T B\left(\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right)\right\|^2 \end{aligned} \tag{3.2.4}$$

Proof. The classical Augmented Lagrangian has form

$$L_\beta(x^{\nu},\mu^{\nu})=f(x^{\nu})+\langle\mu^{\nu},Ax^{\nu}\rangle+\frac{\beta}{2}\left\|Ax^{\nu}\right\|^2$$

Considering (3.1.7), the μ-variation of L_β is given by

$$\begin{aligned} L_\beta(x^{\nu+1},\mu^{\nu+1})-L_\beta(x^{\nu+1},\mu^{\nu}) &= \left\langle\mu^{\nu+1}-\mu^{\nu},Ax^{\nu+1}\right\rangle \\ &= \frac{1}{\beta}\|\mu^{\nu+1}-\mu^{\nu}\|^2 \end{aligned} \tag{3.2.5}$$

The overall L_β variation will be:

$$\begin{aligned}
L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^\nu,\mu^\nu) =& L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^{\nu+1},\mu^\nu) \\
&+ L_\beta(x^{\nu+1},\mu^\nu) - L_\beta(x^\nu,\mu^\nu) \\
\leq& L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^{\nu+1},\mu^\nu) \\
&+ L_\beta(x^{\nu+1},\mu^\nu) + \frac{\beta}{2}\left\|x^{\nu+1}-x^\nu\right\|^2_{B^TB} \\
&- L_\beta(x^\nu,\mu^\nu)
\end{aligned}$$

With the use of the equation (3.2.5), and the fact that $L_\beta(x,\mu^\nu) + (\beta/2)\|x - x^\nu\|^2_{B^TB}$ is strongly convex we obtain

$$\begin{aligned}
L_\beta(x^{\nu+1},\mu^{\nu+1}) \leq L_\beta(x^\nu,\mu^\nu) &+ \frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^\nu\right\|^2 - \frac{\gamma}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 \\
&+ \left\langle \nabla_x L_\beta(x^{\nu+1},\mu^\nu) + \beta B^TB(x^{\nu+1}-x^\nu), x^{\nu+1}-x^\nu\right\rangle
\end{aligned} \tag{3.2.6}$$

From the discrete-time update of (3.1.9) we obtain:

$$\begin{aligned}
\nabla_x L_\beta(x^{\nu+1},\mu^\nu) + \beta B^TB(x^{\nu+1}-x^\nu) &= \nabla f(x^{\nu+1}) + A^T\mu^\nu + \beta A^TAx^{\nu+1} + \beta B^TB(x^{\nu+1}-x^\nu) \\
&= \nabla f(x^{\nu+1}) - \nabla f(x^\nu) + 2\beta \mathrm{D}q^{\nu+1}
\end{aligned} \tag{3.2.7}$$

Equation (3.2.7) applied to (3.2.6) yields

$$\begin{aligned}
L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^\nu,\mu^\nu) \leq& \frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^\nu\right\|^2 - \frac{\gamma}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 \\
&+ \langle \nabla f(x^{\nu+1}) - \nabla f(x^\nu), x^{\nu+1}-x^\nu\rangle \\
&+ \left\langle 2\beta \mathrm{D}q^{\nu+1}, x^{\nu+1}-x^\nu\right\rangle
\end{aligned} \tag{3.2.8}$$

Applying Young's Inequality (Lemma 2.1) we have

$$\left\langle 2\beta \mathrm{D}q^{\nu+1}, x^{\nu+1}-x^\nu\right\rangle \leq \frac{\epsilon}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}q^{\nu+1}\right\|^2 \tag{3.2.9}$$

for all $\epsilon > 0$. Taking into account (3.2.9) in (3.2.8) and Assumption 1 we obtain

$$\begin{aligned}
L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^\nu,\mu^\nu) \leq& \frac{1}{\beta}\left\|\mu^{\nu+1}-\mu^\nu\right\|^2 - \frac{\gamma-2\mathrm{L}}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 \\
&+ \frac{\epsilon}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}q^{\nu+1}\right\|^2
\end{aligned} \tag{3.2.10}$$

Lastly, inequality (3.2.1) is applied to (3.2.10) resulting in

$$\begin{aligned} L_\beta(x^{\nu+1},\mu^{\nu+1}) - L_\beta(x^\nu,\mu^\nu) \leq & -\frac{\gamma - 2\mathrm{L} - \epsilon}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}q^{\nu+1}\right\|^2 \\ & + \frac{3\mathrm{L}^2}{\beta\underline{\sigma}^2(A)}\left\|x^\nu - x^{\nu-1}\right\|^2 \\ & + \frac{12\beta}{\underline{\sigma}^2(A)}\left\|\,\mathrm{D}\left(q^{\nu+1}-q^\nu\right)\right\|^2 \\ & + \frac{3\beta}{\underline{\sigma}^2(A)}\left\|B^TB\left(\left(x^{\nu+1}-x^\nu\right)-\left(x^\nu-x^{\nu-1}\right)\right)\right\|^2 \end{aligned} \tag{3.2.11}$$

which completes the proof of the lemma. □

From Lemma 3.2, can be observed that L_β is not decreasing when the iteration increases. This is clearly shown by the terms that have positive sign. We also consider the variation of the nonnegative term

$$S_\beta(x^{n+1},x^\nu) := \frac{\beta}{2}\left(\left\|Ax^{\nu+1}\right\|^2 + \left\|x^{\nu+1}-x^\nu\right\|^2_{B^TB}\right)$$

The following lemma illustrates the variation of S.

Lemma 3.3. Suppose that assumption A.1 is satisfied. Then, the following inequality holds true

$$\begin{aligned} S_\beta(x^{\nu+1},x^\nu) \leq & S_\beta(x^\nu,x^{\nu-1}) - \frac{\beta}{2}\left\|A\left(x^{\nu+1}-x^\nu\right)\right\|^2 - \frac{\beta}{2}\left\|\left(x^{\nu+1}-x^\nu\right)-\left(x^\nu-x^{\nu-1}\right)\right\|^2_{B^TB} \\ & + \frac{\mathrm{L}+\epsilon}{2}\left\|x^{\nu+1}-x^\nu\right\|^2 + \frac{\mathrm{L}}{2}\left\|x^\nu-x^{\nu-1}\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}\left(q^{\nu+1}-q^\nu\right)\right\|^2 \end{aligned} \tag{3.2.12}$$

Proof. From the discrete-time update (3.1.9), we obtain:

$$\left\langle \nabla f(x^\nu) + A^T\mu^\nu + \beta A^TAx^{\nu+1} + \beta B^TB(x^{\nu+1}-x^\nu) - 2\beta\mathrm{D}q^{\nu+1}, x^{\nu+1}-x\right\rangle \leq 0, \forall x \in \mathrm{R}^Q$$

$$\left\langle \nabla f(x^{\nu-1}) + A^T\mu^{\nu-1} + \beta A^TAx^{\nu} + \beta B^TB(x^{\nu}-x^{\nu-1}) - 2\beta\mathrm{D}q^{\nu}, x^{\nu}-x\right\rangle \leq 0, \forall x \in \mathrm{R}^Q$$

Plugging $x = x^\nu$ into the first inequality and $x = x^{\nu+1}$ into the second, adding the resulting inequalities and utilizing the dual variable update, we obtain,

$$\begin{aligned} \Big\langle \nabla f(x^\nu) - \nabla f(x^{\nu-1}) + A^T(\mu^{\nu+1}-\mu^\nu) + \beta B^TB\left(\left(x^{\nu+1}-x^\nu\right)-\left(x^\nu-x^{\nu-1}\right)\right) \\ -2\beta\mathrm{D}(q^{\nu+1}-q^\nu), x^{\nu+1}-x^\nu\Big\rangle \leq 0 \end{aligned}$$

Equivalently, we have

$$\begin{aligned} \left\langle A^T(\mu^{\nu+1}-\mu^\nu), x^{\nu+1}-x^\nu \right. \leq & -\left\langle \nabla f(x^\nu) - \nabla f(x^{\nu-1}), x^{\nu+1}-x^\nu\right\rangle \\ & -\left\langle \beta B^TB\left(\left(x^{\nu+1}-x^\nu\right)-\left(x^\nu-x^{\nu-1}\right)\right) - 2\beta\mathrm{D}(q^{\nu+1}-q^\nu), x^{\nu+1}-x^\nu\right\rangle \end{aligned} \tag{3.2.13}$$

The next step of the proof is to bound both sides of the inequality. For the lhs of (3.2.13) it can be expressed as

$$\begin{aligned}\left\langle A^T(\mu^{\nu+1}-\mu^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle &= \left\langle \beta A^T A x^{\nu+1}, x^{\nu+1}-x^{\nu}\right\rangle \\ &= \left\langle \beta A x^{\nu+1}, A x^{\nu+1}-A x^{\nu}\right\rangle \\ &= \beta \left\|A x^{\nu+1}\right\|^2 - \beta \left\langle A x^{\nu+1}, A x^{\nu}\right\rangle \\ &= \frac{\beta}{2}\left(\left\|A x^{\nu+1}\right\|^2 - \left\|A x^{\nu}\right\|^2 + \left\|A(x^{\nu+1}-x^{\nu})\right\|^2\right)\end{aligned}$$

Therefore, the lhs will be:

$$\left\langle A^T(\mu^{\nu+1}-\mu^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle = \frac{\beta}{2}\left(\left\|A x^{\nu+1}\right\|^2 - \left\|A x^{\nu}\right\|^2 + \left\|A(x^{\nu+1}-x^{\nu})\right\|^2\right) \tag{3.2.14}$$

Secondly, the next step is to bound the rhs of the inequality (3.2.13).

$$\begin{aligned}&-\left\langle \nabla f(x^{\nu}) - \nabla f(x^{\nu-1}) + \beta B^T B\left(\left(x^{\nu+1}-x^{\nu}\right) - \left(x^{\nu}-x^{\nu-1}\right)\right) - 2\beta \mathrm{D}(q^{\nu+1}-q^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle \\ &= -\left\langle \nabla f(x^{\nu}) - \nabla f(x^{\nu-1}), x^{\nu+1}-x^{\nu}\right\rangle - \beta\left\langle B^T B\left(\left(x^{\nu+1}-x^{\nu}\right)\right), x^{\nu+1}-x^{\nu}\right\rangle \\ &\quad + \beta\left\langle B^T B(x^{\nu}-x^{\nu-1}), x^{\nu+1}-x^{\nu}\right\rangle + 2\beta\left\langle \mathrm{D}(q^{\nu+1}-q^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle \\ &\le \mathrm{L}\left\|x^{\nu}-x^{\nu-1}\right\|\left\|x^{\nu+1}-x^{\nu}\right\| - \beta\left\|x^{\nu+1}-x^{\nu}\right\|^2_{B^TB} \\ &+ \beta\left\langle B^T B(x^{\nu}-x^{\nu-1}), x^{\nu+1}-x^{\nu}\right\rangle + 2\beta\left\langle \mathrm{D}(q^{\nu+1}-q^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle \\ &\le \frac{\mathrm{L}}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2 + \frac{\mathrm{L}}{2}\left\|x^{\nu+1}-x^{v}\right\|^2 \\ &+ \frac{\beta}{2}\left(\left\|x^{\nu}-x^{\nu-1}\right\|^2_{B^TB} - \left\|x^{\nu+1}-x^{\nu}\right\|^2_{B^TB} - \left\|(x^{\nu}-x^{\nu-1}) - (x^{\nu+1}-x^{\nu})\right\|^2_{B^TB}\right) \\ &+ \frac{\epsilon}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}(q^{\nu+1}-q^{\nu})\right\|^2\end{aligned}$$

Thus, the rhs will be:

$$\begin{aligned}&-\left\langle \nabla f(x^{\nu}) - \nabla f(x^{\nu-1}) + \beta B^T B\left(\left(x^{\nu+1}-x^{\nu}\right) - (x^{\nu}-x^{\nu-1})\right) - 2\beta \mathrm{D}(q^{\nu+1}-q^{\nu}), x^{\nu+1}-x^{\nu}\right\rangle \\ &\le \frac{\mathrm{L}}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2 + \frac{\mathrm{L}}{2}\left\|x^{\nu+1}-x^{\nu}\right\| + \frac{\beta}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2_{B^TB} - \frac{\beta}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2_{B^TB} \\ &- \frac{\beta}{2}\left\|(x^{\nu}-x^{\nu-1}) - (x^{\nu+1}-x^{\nu})\right\|^2_{B^TB} + \frac{\epsilon}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}(q^{\nu+1}-q^{\nu})\right\|^2\end{aligned} \tag{3.2.15}$$

Combining the inequalities (3.2.14) and (3.2.15),

$$\begin{aligned}&\frac{\beta}{2}\left(\left\|A x^{\nu+1}\right\|^2 - \left\|A x^{\nu}\right\|^2 + \left\|A(x^{\nu+1}-x^{\nu})\right\|^2\right) \\ &\le \frac{\mathrm{L}}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2 + \frac{\mathrm{L}}{2}\left\|x^{\nu+1}-x^{\nu}\right\| + \frac{\beta}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2_{B^TB} - \frac{\beta}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2_{B^TB} \\ &- \frac{\beta}{2}\left\|(x^{\nu}-x^{\nu-1}) - (x^{\nu+1}-x^{\nu})\right\|^2_{B^TB} + \frac{\epsilon}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2 + \frac{2\beta^2}{\epsilon}\left\|\mathrm{D}(q^{\nu+1}-q^{\nu})\right\|^2\end{aligned}$$

Rearranging the above inequality, the following result has been proven of (3.2.12).

$$\begin{aligned}\frac{\beta}{2}\left(\left\|Ax^{\nu+1}\right\|^2+\left\|x^{\nu+1}-x^{\nu}\right\|^2_{B^TB}\right)\leq&\frac{\beta}{2}\left\|Ax^{\nu}\right\|^2+\frac{\beta}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2_{B^TB}\\&-\frac{\beta}{2}\left\|A\left(x^{\nu+1}-x^{\nu}\right)\right\|^2+\frac{\mathrm{L}+\epsilon}{2}\left\|x^{\nu+1}-x^{\nu}\right\|^2\\&+\frac{\mathrm{L}}{2}\left\|x^{\nu}-x^{\nu-1}\right\|^2+\frac{2\beta^2}{\epsilon}\left\|\mathrm{D}\left(q^{\nu+1}-q^{\nu}\right)\right\|^2\\&-\frac{\beta}{2}\left\|\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right\|^2_{B^TB}\end{aligned}$$

□

We can now combine $L_\beta(x^{\nu+1},\mu^{\nu+1})$ and $S_\beta(x^{\nu+1},x^{\nu})$ to obtain an overall Lypunov-like function. Firstly, let a constant $c>0$. The new potential function is given by

$$P_{c,\beta}\left(x^{\nu+1},x^{\nu},\mu^{\nu}\right)=L_\beta\left(x^{\nu+1},\mu^{\nu+1}\right)+cS_\beta(x^{\nu+1},x^{\nu}) \tag{3.2.16}$$

Lemma 3.4. Consider N agents with outputs $x_1,x_2,...,x_N\in R^m$. Suppose that the assumptions made on Lemmas 3.1 and 3.3 are satisfied. Then, for the error variables $q_1,..,q_N$ defined by it holds true that the following has been proven

$$\begin{aligned}P_{c,\beta}\left(x^{\nu+1},x^{\nu},\mu^{\nu+1}\right)\leq&P_{c,\beta}\left(x^{\nu},x^{\nu-1},\mu^{\nu}\right)-\left(\frac{\gamma-2\mathrm{L}-\epsilon-c(\mathrm{L}+\epsilon)}{2}\right)\left\|x^{\nu+1}-x^{\nu}\right\|^2\\&+\left(\frac{3\mathrm{L}^2}{\beta\underline{\sigma}^2(A)}+\frac{c\mathrm{L}}{2}\right)\left\|x^{\nu}-x^{\nu-1}\right\|^2\\&-\left(\frac{c\beta}{2}-\frac{3\left\|B^TB\right\|\beta}{\underline{\sigma}^2(A)}\right)\left\|\left(x^{\nu+1}-x^{\nu}\right)-\left(x^{\nu}-x^{\nu-1}\right)\right\|^2_{B^TB}\\&+\frac{2\beta^2}{\epsilon}\left\|\mathrm{D}q^{\nu+1}\right\|^2+\left(\frac{2\beta^2c}{\epsilon}+\frac{12\beta}{\underline{\sigma}^2(A)}\right)\left\|\mathrm{D}(q^{\nu+1}-q^{\nu})\right\|^2\end{aligned} \tag{3.2.17}$$

Proof. Multiplying both sides of (3.2.12) by c and summing with (3.2.4) the above inequality is directly obtained. □

As it can be seen from the previous lemma, there are positive terms in the rhs of the inequality, which involve the term $\left\|x^{\nu}-x^{\nu-1}\right\|^2$ as well as the error terms with the q variables.

Taking the summation over all iterations from 1 up to $\nu+1$ we have:

$$\begin{aligned} P_{c,\beta}\left(x^{\nu+1},x^{\nu},\mu^{\nu+1}\right) \leq P_{c,\beta}\left(x^{1},x^{0},\mu^{1}\right) &- \left(\frac{\gamma-2\mathrm{L}-\epsilon-c(\mathrm{L}+\epsilon)}{2}\right)\left\|x^{\nu+1}-x^{\nu}\right\|^{2} \\ &- \left(\frac{\gamma-2\mathrm{L}-\epsilon-c(\mathrm{L}+\epsilon)}{2}-\frac{3\mathrm{L}^{2}}{\beta\underline{\sigma}^{2}(A)}-\frac{c\mathrm{L}}{2}\right)\sum_{\ell=1}^{\nu-1}\left\|x^{\ell+1}-x^{\ell}\right\|^{2} \\ &- \left(\frac{c\beta}{2}-\frac{3\beta\|B^{T}B\|}{\underline{\sigma}^{2}(A)}\right)\sum_{\ell=1}^{\nu}\left\|\left(x^{\ell+1}-x^{\ell}\right)-\left(x^{\ell}-x^{\ell-1}\right)\right\|_{B^{T}B}^{2} \\ &+ \frac{2\beta^{2}}{\epsilon}\sum_{\ell=1}^{\nu}\left\|\mathrm{D}q^{\ell+1}\right\|^{2}+\left(\frac{2\beta^{2}c}{\epsilon}+\frac{12\beta}{\underline{\sigma}^{2}(A)}\right)\sum_{\ell=1}^{\nu}\left\|\mathrm{D}(q^{\ell+1}-q^{\ell})\right\|^{2} \end{aligned} \tag{3.2.18}$$

It is clear that the two design parameters must have large values to ensure that the respective terms in the Lyapunov variation are nonpositive. Also, the error variable terms in (3.2.18) have a destabilizing effect. We claim that if

$$\sum_{\ell=0}^{\infty}\left\|q_{i}^{\ell}\right\|^{2}<\infty,\quad \sum_{\ell=0}^{\infty}\left\|q_{i}^{\ell+1}-q_{i}^{\ell}\right\|^{2}<\infty \tag{3.2.19}$$

then boundedness of $P_{c,\beta}$ can be proved assuming sufficiently large c,β. Moreover, since we will be working with the continuous variable q, it is crucial to find conditions on $q(t)$ that guarantee the ℓ_2 property of the samples. In this respect, it is proven that the condition (3.2.19) will be secured through the design of robust distributed control laws that will ensure that

$$\int_{0}^{\infty}\|q_{i}(s)\|\,ds<\infty,\qquad \int_{0}^{\infty}\|q_{i}(s)\|^{2}\,ds<\infty \tag{3.2.20}$$

Thus, $q_i(t)\in\mathcal{L}_1\cap\mathcal{L}_2 \Rightarrow q_i^k\in\ell_2$.

From the analysis of Lemma 3.4 it is easy to see that as long as c and β are chosen large enough, then the respective terms in the variation of the potential function will be nonpositive. A sufficient condition for the constant c is derived as:

$$\frac{c\beta}{2}-\frac{3\beta\|B^{T}B\|}{\underline{\sigma}^{2}(A)}>0$$

$$\Rightarrow c>\frac{6\|B^{T}B\|}{\underline{\sigma}^{2}(A)}$$

Now, a sufficient condition for the constant c will be

$$c\geq\max\left\{\frac{\delta}{\mathrm{L}},\frac{6\|B^{T}B\|}{\underline{\sigma}^{2}(A)}\right\} \tag{3.2.21}$$

As a note, the term δ/L in the max operator is going to be used later in the analysis for the convergence of the algorithm. Thus, this is the lower bound of the constant c. In addition, the proven bound is independent of the penalty parameter β. Second, for any given constant

c, the parameter β must satisfy the condition

$$\frac{\gamma - 2\mathrm{L} - \epsilon - c(\mathrm{L}+\epsilon)}{2} - \frac{3\mathrm{L}^2}{\beta\underline{\sigma}^2(A)} - \frac{c\mathrm{L}}{2} > 0$$

$$\Rightarrow \gamma > 2\mathrm{L} + \epsilon + c(\mathrm{L}+\epsilon) + \frac{6\mathrm{L}^2}{\beta\underline{\sigma}^2(A)} + c\mathrm{L}$$

$$\Rightarrow \beta > \epsilon + c(\mathrm{L}+\epsilon) + \frac{6\mathrm{L}^2}{\beta\underline{\sigma}^2(A)} + c\mathrm{L}$$

$$\Rightarrow \beta^2 - (2\mathrm{L}c + (1+c)\epsilon)\beta - \frac{6\mathrm{L}^2}{\underline{\sigma}^2(A)} > 0$$

So, in order to have a large sufficient value, only the positive root of the above polynomial will be used. Therefore,

$$\beta > \frac{1}{2}\left(2\mathrm{L}c + (1+c)\,\epsilon + \sqrt{(2\mathrm{L}c + (1+c)\,\epsilon)^2 + \frac{24\mathrm{L}^2}{\underline{\sigma}^2(A)}}\right) \tag{3.2.22}$$

It is clear that combining the bounds of β and c, the result is that $\beta > \delta$. It is concluded that if both bounds (3.2.21) and (3.2.22) are satisfied, then the corresponding terms in the potential function $P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1})$ variation will be nonpositive. With the use of (3.2.18) and the particular choice of c and β, then the constructed potential function will be lower bounded.

Lemma 3.5. Suppose that the assumptions A.1 - A.3 are satisfied, and β, c are chosen according to (3.2.21) and (3.2.22). If, for $q_i(t)$ defined by (3.1.1), the error variable sample sequence $\{q_i^k\}$ is ℓ_2 then the sequence $\{P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right)\}$ is uniformly bounded.

Proof. To prove lower boundedness, first will be proven the summation of the potential function that is lower bounded. In order to prove it, assumption A.2 is going to be utilized. First of all the augmented Lagrangian is expressed as follow

$$\begin{aligned}
L_\beta(x^{\nu+1}, \mu^{\nu+1}) &= f(x^{\nu+1}) + \left\langle \mu^{\nu+1}, Ax^{\nu+1}\right\rangle + \frac{\beta}{2}\left\|Ax^{\nu+1}\right\|^2 \\
&= f(x^{\nu+1}) + \frac{1}{\beta}\left\langle \mu^{\nu+1}, \mu^{\nu+1} - \mu^{\nu}\right\rangle + \frac{\beta}{2}\left\|Ax^{\nu+1}\right\|^2 \\
&= f(x^{\nu+1}) + \frac{1}{2\beta}\left(\left\|\mu^{\nu+1}\right\|^2 - \left\|\mu^{\nu}\right\|^2 + \left\|\mu^{\nu+1} - \mu^{\nu}\right\|^2\right) \\
&+ \frac{\beta}{2}\left\|Ax^{\nu+1}\right\|^2
\end{aligned}$$

Therefore, summing over $\ell = 1...\nu$, it is obtained

$$\begin{aligned}\sum_{\ell=1}^{\nu} L_\beta(x^{\ell+1}, \mu^{\ell+1}) &= \sum_{\ell=1}^{\nu} \left(f(x^{\ell+1}) + \frac{1}{2\beta} \left\|\mu^{\ell+1} - \mu^{\ell}\right\|^2 + \frac{\beta}{2} \left\|Ax^{\ell+1}\right\|^2 \right) \\ &+ \frac{1}{2\beta} \left(\sum_{\ell=1}^{\nu} \left\|\mu^{\ell+1}\right\|^2 - \sum_{\ell=1}^{\nu} \left\|\mu^{\ell}\right\|^2 \right) \\ &= \sum_{\ell=1}^{\nu} \left(f(x^{\ell+1}) + \frac{1}{2\beta} \left\|\mu^{\ell+1} - \mu^{\ell}\right\|^2 + \frac{\beta}{2} \left\|Ax^{\ell+1}\right\|^2 \right) \\ &+ \frac{1}{2\beta} \left[\left\|\mu^{\nu+1}\right\|^2 - \left\|\mu^{1}\right\|^2 \right]\end{aligned}$$

Now A.2 is satisfied and β is chosen according to (3.2.21) and (3.2.22). Then is clear that the above summation is lower bounded since

$$f(x) + \frac{\beta}{2} \|Ax\|^2 \geq f(x) + \frac{\delta}{2} \|Ax\|^2 \geq 0, \ \ \forall x \in \mathrm{R}^Q$$

So, the sum of potential function over ν iterations is

$$\begin{aligned}\sum_{\ell=1}^{\nu} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) &= \sum_{\ell=1}^{\nu} L_\beta\left(x^{\ell+1}, \mu^{\ell+1}\right) + \frac{c\beta}{2} \sum_{\ell=1}^{T} \left(\left\|Ax^{\ell+1}\right\|^2 + \left\|x^{\ell+1} - x^{\ell}\right\|^2_{B^TB} \right) \\ &= \sum_{\ell=1}^{\nu} \left(f(x^{\ell+1}) + \frac{1}{2\beta} \left\|\mu^{\ell+1} - \mu^{\ell}\right\|^2 + \frac{\beta}{2} \left\|Ax^{\ell+1}\right\|^2 \right) \\ &+ \frac{1}{2\beta} \left[\left\|\mu^{\nu+1}\right\|^2 - \left\|\mu^{1}\right\|^2 \right] + \frac{c\beta}{2} \sum_{\ell=1}^{\nu} \left(\left\|Ax^{\ell+1}\right\|^2 + \left\|x^{\ell+1} - x^{\ell}\right\|^2_{B^TB} \right) \\ &\geq \sum_{\ell=1}^{\nu} \left(f(x^{\ell+1}) + \frac{\delta}{2} \left\|Ax^{\ell+1}\right\|^2 + \frac{1}{2\beta} \left\|\mu^{\ell+1} - \mu^{\ell}\right\|^2 \right) \\ &+ \frac{1}{2\beta} \left[\left\|\mu^{\nu+1}\right\|^2 - \left\|\mu^{1}\right\|^2 \right] \\ &+ \frac{c\beta}{2} \sum_{\ell=1}^{\nu} \left(\left\|Ax^{\ell+1}\right\|^2 + \left\|x^{\ell+1} - x^{\ell}\right\|^2_{B^TB} \right) \geq 0 \\ &\Rightarrow \sum_{\ell=1}^{\nu} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) > -\infty, \ \ \forall \nu > 0.\end{aligned}$$

The summation of the potential function has been proven that is lower bounded. The next step is to ensure that the potential function is upper and lower bounded. Utilizing the fact that $q_i \in \ell_2$ the upper bound is obtained from a direct application of (3.2.18) as follows:

$$P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1}) \leq P_{c,\beta}(x^1, x^0, \mu^1) + \gamma_q \sum_{\ell=1}^{\nu} \|q^{\ell+1}\|^2$$

Therefore, if $q^k \in \ell_2$ then $P_{c,\beta}$ is upper bounded. We will now prove that it is also lower

bounded. Assume the opposite, i.e. for any constant $N_0 > 0$ there exists a strictly increasing subsequence $\{\ell_k\} \subset N$ with $\lim_{k\to\infty} \ell_k = \infty$ such that $P_{c,\beta}(x^{\ell_k+1}, x^{\ell_k}, \mu^{\ell_k+1}) < -N_0$. From (3.2.17) it can be proved that

$$P_{c,\beta}(x^{\nu_2+1}, x^{\nu_2}, \mu^{\nu_2+1}) \leq P_{c,\beta}(x^{\nu_1}, x^{\nu_1-1}, \mu^{\nu_1}) + \gamma_q \sum_{\ell=\nu_1}^{\nu_2} \|q^{\ell+1}\|^2$$

We can now prove that there exists some $\kappa \in N$ such that $P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1}) \leq 0$ for all $\nu \geq \kappa$. Note that the ℓ_2 property of q^ν implies that $\lim_{\nu\to\infty} \sum_{\ell=\nu}^{\infty} \|q^\ell\|^2 = 0$. Hence, there exists κ such that

$$\sum_{\ell=\nu}^{\infty} \|q^\ell\|^2 \leq \frac{N_0}{\gamma_q}, \quad \forall \nu \geq \kappa$$

Let ℓ_k^* be the smallest element of $\{\ell_k\}$ which is larger than κ. Then, for all times ν that do not belong to the subsequence $\{\ell_k\}$ and are larger than ℓ_k^* we have

$$\begin{aligned} P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1}) \leq & P_{c,\beta}(x^{\ell_k^*+1}, x^{\ell_k^*}, \mu^{\ell_k^*+1}) + \gamma_q \sum_{\ell=\ell_k^*}^{\nu} \|q^{\ell+1}\|^2 \\ \leq & -N_0 + \gamma_q \sum_{\ell=\ell_k^*}^{\infty} \|q^{\ell+1}\|^2 \leq 0 \end{aligned}$$

As a result, we have that

$$\begin{aligned} \sum_{\ell=1}^{\infty} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) = & \sum_{\ell\in\{\ell_k\}} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) + \sum_{\ell>\ell_k^* \wedge \ell\notin\{\ell_k\}} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) \\ & + \sum_{\ell\in\{1,\ldots,\ell_k^*-1\}\wedge\ell\notin\{\ell_k\}} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) \end{aligned}$$

The first term in the rhs of the above equation is equal to $-\infty$ since each term within the sum is less than or equal to $-N_0$ and there are infinite such terms (by assumption). The second term is nonpositive since each of the terms in the sum is nonpositive. The third term is upper bounded since it includes a finite number of upper bounded terms. As a result $\sum_{\ell=1}^{\infty} P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right) = -\infty$ which yields the desired contradiction. This concludes the proof of the lower boundedness property of $P_{c,\beta}\left(x^{\ell+1}, x^{\ell}, \mu^{\ell+1}\right)$. □

All the above results are going to be used for proving the main theorem of this thesis. The next and final step is to evaluate the convergence rate of the algorithm. Let $Q(x^{\nu+1}, \mu^{\nu+1})$ be the optimality gap of problem (2.5.6), which is given by:

$$Q(x^{\nu+1}, \mu^{\nu+1}) := \left\|\nabla_x L_\beta(x^{\nu+1}, \mu^{\nu+1})\right\|^2 + \beta \left\|Ax^{\nu+1}\right\|^2 \tag{3.2.23}$$

Clearly, it can be seen that when equation (3.2.23) tends to zero, the limit point $(x_\nu^\bullet, \mu_\nu^\bullet)$ is a

KKT point of the problem (2.5.6). The optimal solution is summarized from the equations (3.2.24) as shown below:

$$\begin{aligned} \nabla f(x_\nu^\bullet) + A^T \mu_\nu^\bullet &= 0 \\ A x_\nu^\bullet &= 0 \end{aligned} \tag{3.2.24}$$

Since $A(\mathbf{1}_N \otimes I) = 0$, utilizing the above equations, one can prove that:

$$\left(1_N^T \otimes I_n\right) \nabla f(x_\nu^\bullet) = 0 \Rightarrow \sum_{i=1}^{N} \nabla f_i(x_\nu^\bullet) = 0$$

In order to reach the optimal condition, the consensus must be achieved, i.e. $Ax^{\nu+1} \to 0$. Therefore, a necessary condition is that the sequence of the dual variable variations $\mu^{\nu+1} - \mu^\nu$ must converge to 0, which is achieved when the sequence of the primal variable converges. The first condition of (3.2.24) can be proved from the optimality gap when it tends to zero. So,

$$\lim_{\nu\to\infty} \nabla f(x^\nu) + A^T \mu^\nu + \beta A^T A x^{\nu+1} = \nabla f(x_\nu^\bullet) + A^T \mu_\nu^\bullet = 0$$

The next and final step of the analysis is to ensure the convergence of the algorithm through the optimality gap $Q(\cdot)$. The optimality gap tends to zero in a sublinear manner. This is presented in the following theorem.

Theorem 3.1. Suppose that the main assumptions A.1 - A.4 are true and satisfied. In addition, suppose that the parameters c and β are chosen such as are presented in (3.2.21) and (3.2.22), respectively. Assume also that, for the error variables $q_i(t)$ defined by (3.1.1), the sample sequence $\{q^k\}_{k=1}^{\infty}$ is ℓ_2. Then, the following claims hold:

- The constraint is satisfied in the limit, i.e.

$$\lim_{\nu\to\infty} \mu^{\nu+1} - \mu^\nu = 0$$

$$\lim_{\nu\to\infty} Ax^{\nu+1} = 0$$

$$\lim_{\nu\to\infty} x^{\nu+1} - x^\nu = 0$$

$$\lim_{t\to\infty} x(t) = x^*$$

- Furthermore, if $\|f(x)\|$ is bounded for all $x \in \mathrm{R}^Q$, then the sequence $\{\mu^\ell\}_{\ell=1}^{\nu+1}$ is also bounded; If $f(x) + \frac{\beta}{2}\|Ax\|^2$ is coercive, then the sequence $\{x^\ell\}_{\ell=1}^{\nu+1}$ is bounded.

- Every limit point which is produced from the iterates $\{x^\ell, \mu^\ell\}_{\ell=1}^{\nu+1}$, which are generated from the time discrete update of remark 3.3, converges to a stationary point of problem (2.5.6). In addition, $Q(x^{\nu+1}\, \mu^\nu) \to 0$.

Proof.

1. From (3.2.18) utilizing the fact that $P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1})$ is bounded and $q^k \in \ell_2$ we obtain that $\sum_{\nu=1}^{\infty} \|x^{\nu+1} - x^{\nu}\|^2 < \infty$, $\sum_{\nu=1}^{\infty} \|(x^{\nu+1} - x^{\nu}) - (x^{\nu} - x^{\nu-1})\|_{B^TB}^2 < \infty$ which in turn implies that $x^{\nu+1} - x^{\nu} \to 0$, $B^TB((x^{\nu+1} - x^{\nu}) - (x^{\nu} - x^{\nu-1})) \to 0$. In addition, with the use of lemma 3.1 it is concluded that $\|\mu^{\nu+1} - \mu^{\nu}\|^2 \to 0$. So the dual variable variation also converges, $\mu^{\nu+1} - \mu^{\nu} \to 0$. This in turn yields that the constraint is satisfied in the limit, $Ax^{\nu+1} \to 0$, and consensus between the agents is achieved.

2. The discrete time update has form:

$$\nabla f(x^{\nu}) + A^T\mu^{\nu} + \beta A^TAx^{\nu+1} + \beta B^TB(x^{\nu+1} - x^{\nu}) - 2\beta \mathrm{D}q^{\nu+1} = 0$$

As a first step, the sequence $\{\mu^{\ell}\}_{\ell=1}^{\nu+1}$ must be bounded, if $\nabla f(x^{\nu})$ is bounded. This statement is true, since from step 1 has been proved that $x^{\nu+1} - x^{\nu} \to 0$ and $Ax^{\nu+1} \to 0$. Then, from the assumption that $\nabla f(x)$ is bounded for any $x \in \mathbf{R}^Q$ and the fact that the dual variable lies into the image space of the incidence matrix of the system, the sequence $\{\mu^{\ell}\}_{\ell=1}^{\nu+1}$ is bounded.

If $f(x) + \frac{\beta}{2}\|Ax\|^2$ is coercive, then it will ensure that the primal variable is bounded. Utilizing the potential function, we obtain:

$$\begin{aligned} P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1}) =& L_{\beta}(x^{\nu+1}, \mu^{\nu+1}) + cS_{\beta}(x^{\nu+1}, x^{\nu}) \\ &= f(x^{\nu+1}) + \frac{1}{2\beta}\left(\|\mu^{\nu+1}\|^2 - \|\mu^{\nu}\|^2 + \|\mu^{\nu+1} - \mu^{\nu}\|^2\right) \\ &+ \frac{c\beta}{2}\left(\|Ax^{\nu+1}\|^2 + \|x^{\nu+1} - x^{\nu}\|_{B^TB}^2\right) \end{aligned}$$

From lemma 3.5 $P_{c,\beta}$ has been proven to be both upper and lower bounded. Suppose that the sequence $\{x^{\ell}\}_{\ell=1}^{\nu+1}$ is not bounded. Let a domain $\mathcal{K}$ an infinite subset of the iteration index in which $\lim_{\nu\in\mathcal{K}} x^{\nu} = \infty$. Now, this limit is going to be passed to the potential function. So, in this infinite subset the potential function has limit

$$\lim_{\nu\in\mathcal{K}} P_{c,\beta}(x^{\nu+1}, x^{\nu}, \mu^{\nu+1}) = \lim_{\nu\in\mathcal{K}} f(x^{\nu+1}) + \frac{c\beta + \beta}{2}\left\|Ax^{\nu+1}\right\|^2 = \infty$$

Some terms are zero, since $\mu^{\nu+1} \to \mu^{\nu}$ and $x^{\nu+1} \to x^{\nu}$. In addition, the last result of the limit comes from the coerciveness assumption. However, this contradicts the fact that the potential function is upper bounded. So, the sequence $\{x^{\ell}\}_{\ell=1}^{\nu+1}$ is bounded.

□

With the results of theorem 3.1 it is proven that the discrete form of the algorithm can be applied in a real network of agents. However, depended on the properties of the cost functions the sequence $\{(\mu^\ell, x^\ell)\}_{\ell=1}^{\nu+1}$ may or may not be bounded. On the contrary, the optimality gap converges to zero in a sublinear manner, thus the agents will always converge to a stationary optimal point. In order to have convergence, the proposed variables must also converge to zero, and be integrable. So, one must design robust control laws that ensure that the error variables tend to zero.

Chapter 4

Distributed Control System Design for NOCGPROX Regulation

In this chapter, the control system is designed in order to regulate the NOCGPROX variables. Firstly, we define the dynamics of every agent. Secondly, we design the control system with the methodlogies of sliding surface and feedback linearization, to ensure robustness and convergence of the proposed variables to zero.

4.1 Heterogeneous Multi-Agent Dynamics

Consider a group of N heterogeneous agents, which are described by the following dynamics

$$M_i(p_i)\ddot{p}_i + C_i(p_i, \dot{p}_i)\dot{p}_i + G_i(p_i) + F_i(p_i)\dot{p}_i = u_i + d_i(t) \tag{4.1.1}$$

where $p_i \in \mathbf{R}^{nN}$ is the position vector of the $i-$th agents, $\dot{p}_i \in \mathbf{R}^{nN}$ is the velocity vector and $\ddot{p}_i \in \mathrm{R}^{nN}$ is the acceleration vector. The matrix $M_i(p_i)$ is the mass matrix, the matrix $C(p_i, \dot{p}_i)$ is the Coriolis - Centrifugal Forces matrix, $G_i(p_i)$ is the Weight Force matrix and $D(p_i)$ is the viscous friction parameter vector. The vector u_i is the torque input inside the system and $d_i(t)$ is the bounded disturbances that occur in the system. In a more compact form, the above system can be described in a state space form as

$$\begin{bmatrix} \dot{p}_i \\ \ddot{p}_i \end{bmatrix} = \begin{bmatrix} \dot{p}_i \\ -M_i^{-1}(p_i)[C_i(p_i, \dot{p}_i)\dot{p}_i + G_i(p_i) + D_i(p_i)\dot{p}_i] \end{bmatrix} + \begin{bmatrix} 0 \\ M_i^{-1}(p_i) \end{bmatrix} (u_i + d_i(t)) \tag{4.1.2}$$

where $F_i(p_i, \dot{p}_i)$ and $g_i(p_i)$ are the local continuous Lipschitz vector fields, with the output be $x_i := p_i$. So the first order dynamic equation will be

$$\dot{p} = F(p, \dot{p}) + g(p)(u + d(t)) \tag{4.1.3}$$

After the definition of the dynamics for the agents, it is shown below some basic assumptions for the system. Firstly, the disturbances are uniformly bounded as shown in A.5.

Assumption 6. There exists constants $d_i > 0$ such that $\|d_i(t)\| \leq D_i$ for every $t \geq 0, i = 1, ..., N$.

Below, are shown the basic assumptions for the matrices of the dynamic for every agent.

Assumption 7. The matrices M_i are symmetrical and positive defined. There exists positive constants $m_{i2} \geq m_{i1} > 0$ such that $\lambda_{max}(M_i(p_i)) \leq m_{i2}$ and $\lambda_{min}(M_i(p_i)) \geq m_{i1}$, for every $p_i \in \mathbf{R}^n, i = 1, ..., N$. Thus,

$$m_{i1} I_n \leq M_i(p_i) \leq m_{i2} I_n$$

Assumption 8. The matrix $N(p_i, \dot{p}_i) = \dot{M}(p_i) - 2C(p_i, \dot{p}_i)$ is skew-symmetric, with

$$p_i^T N(p_i, \dot{p}_i) p_i = p_i^T (\dot{M}(p_i) - 2C(p_i, \dot{p}_i))) p_i = 0 \tag{4.1.4}$$

Assumption 9. There exists constants $c_{ij} > 0$, such that $\left\| \frac{\partial M_i(p_i)}{\partial p_{ij}} \right\| \leq c_{ij}$, for every $p_i = [p_{i1}, ..., p_{in}]^T \in \mathbf{R}^n, i = 1, ..., N, j = 1, ..., n$

Remark 4.1. From A.9 it can be shown that

$$\left\| \dot{M}_i(p_i) \right\| = \left\| \sum_{j=1}^{n} \frac{\partial M_i(p_i)}{\partial p_{ij}} \dot{p}_{ij} \right\| \leq \sum_{j=1}^{n} \left\| \frac{\partial M_i(p_i)}{\partial p_{ij}} \right\| |\dot{p}_{ij}| \leq \sum_{j=1}^{n} c_{ij} |\dot{p}_{ij}| \leq c_{M_i} \|\dot{p}_i\| \tag{4.1.5}$$

where $c_{M_i} > 0$.

Now that all assumptions for the system have been defined, the next step is to develop the distributed control system. The control system is based on the methodology of sliding surface and feedback linearization, [64]. The purpose of this controller is to project the error variables inside a surface manifold. This will ensure robstuness, while the signals will be secured from external disturbances. The purpose of feedback linearization is to ensure that the nonlinear dynamics will be cancelled from the controller, in order to have a more linear approach. This robust distributed controller will ensure that the NOCGPROX variables will tend to zero and be $\mathcal{L}_1$ in order for the algorithm to converge and achieve optimal consensus.

4.2 Distributed Control System Design for NOCGPROX Variables

Let us introduce the auxiliary variables

$$\omega_i := \dot{q}_i + \vartheta_i q_i \tag{4.2.1}$$

where ϑ_i are static gain parameters. The variables $\omega_i(t)$ create a sliding surface for the NOCGPROX variables, in order to introduce robustness to the system. The next step is to calculate these new variables and insert them inside the dynamics of the agents.

$$\omega_i = \dot{q}_i + \vartheta_i \left[x_i - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) - \left[1 - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right)\right] s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right)\right]$$

Now assume that

$$b_i(t) = \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) + \left[1 - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right)\right] s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right) \tag{4.2.2}$$

then,

$$\omega_i = \dot{p}_i + \vartheta_i p_i - \dot{b}_i(t) - \vartheta_i b_i(t) \tag{4.2.3}$$

When the algorithm is executed, the function $b_i(t)$ generates new sample measurements of the adjacent agents and smoothly incorporates them into the error variables. The next step is to check if the derivative of $b_i(t)$ exists and is continuous in time. Firstly, for time $t \neq kT$ the derivate will be

$$\begin{aligned}\dot{b}_i(t) = &\frac{d}{dt}\rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) + \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) \dot{s}_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) \\ &- \frac{d}{dt}\rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right) - \rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) \dot{s}_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right)\end{aligned}$$

then, the derivate will be

$$\dot{b}_i(t) = \frac{d}{dt}\rho\left(t - \left\lfloor \frac{t}{T} \right\rfloor T\right) \left[s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T\right) - s_i\left(\left\lfloor \frac{t}{T} \right\rfloor T - T\right)\right] \tag{4.2.4}$$

For time periods $t = kT$ the derivative will have value

$$\begin{aligned}\lim_{t \to kT^-} \dot{b}_i(t) &= \left.\frac{d\rho(t)}{dt}\right|_{t=kT} [s_i((k-1)T) - s_i((k-2)T)] = 0 \\ \lim_{t \to kT^+} \dot{b}_i(t) &= \left.\frac{d\rho(t)}{dt}\right|_{t=kT} [s_i(kT) - s_i((k-1)T)] = 0\end{aligned}$$

since $\left.\frac{d\rho}{dt}\right|_{t=kT} = 0$, it can be seen that the function $\dot{b}_i(t)$ is continuous and smooth in its

domain. Furthermore, in the same manner can be proved and for $\ddot{b}_i(t)$. So, for $t \neq kT$ it is

$$\begin{aligned}\ddot{b}_i(t) = &\frac{d^2\rho}{dt^2}\left(t - \left\lfloor\frac{t}{T}\right\rfloor T\right)\left[s_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right) - s_i\left(\left\lfloor\frac{t}{T}\right\rfloor T - T\right)\right] \\ &+ \frac{d\rho}{dt}\left(t - \left\lfloor\frac{t}{T}\right\rfloor T\right)\left[\dot{s}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right) - \dot{s}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T - T\right)\right]\end{aligned}$$

thus, the second derivate of the function will have the form

$$\ddot{b}_i(t) = \frac{d^2\rho}{dt^2}\left(t - \left\lfloor\frac{t}{T}\right\rfloor T\right)\left[s_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right) - s_i\left(\left\lfloor\frac{t}{T}\right\rfloor T - T\right)\right] \tag{4.2.5}$$

For time periods of $t = kT$ the derivative will have the value

$$\begin{aligned}\lim_{t\to kT^-} \ddot{b}_i(t) &= \left.\frac{d^2\rho(t)}{dt^2}\right|_{t=kT}[s_i((k-1)T) - s_i((k-2)T)] = 0 \\ \lim_{t\to kT^+} \ddot{b}_i(t) &= \left.\frac{d^2\rho(t)}{dt^2}\right|_{t=kT}[s_i(kT) - s_i((k-1)T)] = 0\end{aligned}$$

since $\left.\frac{d^r\rho}{dt^r}\right|_{t=kT} = 0$ and the derivative $\ddot{b}_i(t)$ is a well-defined continuous function. As a next step, the derivative of the variable $\omega_i(t)$ is going to be computed as

$$\dot{\omega}_i = \ddot{p}_i + \vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t)) \tag{4.2.6}$$

Now, the equation (4.1.1) will be inserted into equation (4.2.6). So,

$$\dot{\omega}_i = M_i^{-1}(p_i)\left[u_i + d_i(t) - C_i(p_i,\dot{p}_i)\dot{p}_i - G_i(p_i) - F_i(p_i)\dot{p}_i\right] + \vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t)) \tag{4.2.7}$$

Now let be introduced a candidate Lyapunov function for every agent $i \in \mathcal{V}$.

$$V_i := \frac{1}{2}\omega_i^T M_i(p_i)\omega_i \tag{4.2.8}$$

It can be seen that this function is positive definitive as sum of square terms and radially unbounded. The next step is to check the derivative of this function, in order to ensure boundedness of the signals and square integration of the proposed variables. In addition, the control law must have a sign like form in order to ensure and $\mathcal{L}_1$ of the variables. So, the necessary condition will be met in order to prove square integration of the sampled measurements. First, let the expression $Q_i(p_i,\dot{p}_i) = C_i(p_i,\dot{p}_i)\dot{p}_i + G_i(p_i) + F_i(p_i)\dot{p}_i$. The next step is to derive the equation (4.2.8). So,

$$
\begin{aligned}
\dot{V}_i &= \frac{1}{2}\dot{\omega}_i^T M_i(p_i)\omega_i + \frac{1}{2}\omega_i^T M_i(p_i)\dot{\omega}_i + \frac{1}{2}\omega_i^T \dot{M}_i(p_i)\omega_i \\
&= \omega_i^T M_i(p_i)\dot{\omega}_i + \frac{1}{2}\omega_i^T \dot{M}_i(p_i)\omega_i \\
&= \omega_i^T M_i(p_i)\left[M_i^{-1}(p_i)(u_i + d_i(t) - Q_i(p_i,\dot{p}_i)) + \vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right] + \frac{1}{2}\omega_i^T \dot{M}_i(p_i)\omega_i \\
&= \omega_i^T\left[u_i - Q_i(p_i,\dot{p}_i) + M_i(p_i)\left(\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right)\right] + \omega_i^T d_i(t) + \frac{1}{2}\omega_i^T \dot{M}_i(p_i)\omega_i \\
&\leq \omega_i^T\left[u_i - Q_i(p_i,\dot{p}_i)\right] + \|\omega_i\|\,\|M_i(p_i)\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right\| + \|\omega_i\|\,\|d_i(t)\| + \frac{1}{2}\omega_i^T\left\|\dot{M}_i(p_i)\right\|\omega_i
\end{aligned}
$$

now, the assumptions A.6 and A.7 along with remark 4.1 are applied in the above inequality

$$
\begin{aligned}
\dot{V}_i &\leq \omega_i^T\left[u_i - Q_i(p_i,\dot{p}_i)\right] + \|m_{i2}I_n\|\,\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right\| + D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\omega_i^T\omega_i \\
&= \omega_i^T\left[u_i - Q_i(p_i,\dot{p}_i)\right] + m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right\| + D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\,\|\omega_i\|^2
\end{aligned}
$$

The next step is to ensure boundedness of the Lyapunov function. In order to have stabilization in the system, the control signal is chosen as

$$
u_i = Q_i(p_i,\dot{p}_i) - k_{i_1}\omega_i - \left(y_i(\omega_i,\dot{p}_i,\dot{b}_i,\ddot{b}_i) + D_i + k_{i_2}\right)\frac{\omega_i}{\|\omega_i\| + e^{-\left(\lambda_1+\lambda_2 y_i\left(\omega_i,\dot{p}_i,\dot{b}_i,\ddot{b}_i\right)\right)t}} \tag{4.2.9}
$$

where $k_{i_1}, k_{i_2} > 0, \lambda_1, \lambda_2 > 0$, and

$$
y_i(\omega_i,\dot{p}_i,\dot{b}_i,\ddot{b}_i) = m_{i2}\|\vartheta_i\dot{p}_i - (\ddot{b}_i + \vartheta_i\dot{b}_i)\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\|\omega_i\| \tag{4.2.10}
$$

The control signal (4.2.9) guarantees that the sliding manifold ω_i will be bounded and integrable. As a result, the error variable will inherit these propertiesand converge to zero. Thus, proving the necessary condition for achieving optimal consensus between the agents. So, the derivative of the Lyapunov function will become,

$$
\begin{aligned}
\dot{V}_i &\leq \omega_i^T\left[Q_i(p_i,\dot{p}_i) - k_{i_1}\omega_i - (y_i + D_i + k_{i_2})\frac{\omega_i}{\|\omega_i\| + e^{-(\lambda_1+\lambda_2 y_i)t}} - Q_i(p_i,\dot{p}_i)\right] \\
&\quad + m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right\| + D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\,\|\omega_i\|^2 \\
&= -k_{i_1}\|\omega_i\|^2 - (y_i + D_i + k_{i_2})\frac{\|\omega_i\|^2}{\|\omega_i\| + e^{-(\lambda_1+\lambda_2 y_i)t}} + m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t) + \vartheta_i\dot{b}_i(t))\right\| \\
&\quad + D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\,\|\omega_i\|^2
\end{aligned}
$$

Now, the term $\frac{\|\omega_i\|^2}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}}$ can be written as,

$$\begin{aligned}\frac{\|\omega_i\|^2}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}} &= \frac{\|\omega_i\|^2+\|\omega_i\|e^{-(\lambda_1+\lambda_2 y_i)t}}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}} - \frac{\|\omega_i\|e^{-(\lambda_1+\lambda_2 y_i)t}}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}} \\ &= \|\omega_i\|\left(\frac{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}}\right) - \frac{\|\omega_i\|e^{-(\lambda_1+\lambda_2 y_i)t}}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}} \\ &\leq \|\omega_i\| - e^{-(\lambda_1+\lambda_2 y_i)t}\end{aligned}$$

So, the rewritten and simplified expression can be applied to the derivative of the Lyapunov candidate function. Thus,

$$\begin{aligned}\dot{V}_i &\leq -k_{i_1}\|\omega_i\|^2 - (y_i+D_i+k_{i_2})\frac{\|\omega_i\|^2}{\|\omega_i\|+e^{-(\lambda_1+\lambda_2 y_i)t}} + m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t)+\vartheta_i\dot{b}_i(t))\right\| \\ &+ D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\|\omega_i\|^2 \\ &\leq -k_{i_1}\|\omega_i\|^2 - (y_i+D_i+k_{i_2})\left(\|\omega_i\| - e^{-(\lambda_1+\lambda_2 y_i)t}\right) + m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t)+\vartheta_i\dot{b}_i(t))\right\| \\ &+ D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\|\omega_i\|^2 \\ &= -k_{i_1}\|\omega_i\|^2 - (y_i+D_i+k_{i_2})\|\omega_i\| - (y_i+D_i+k_{i_2})e^{-(\lambda_1+\lambda_2 y_i)t} + D_i\|\omega_i\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\|\omega_i\|^2 \\ &+ m_{i2}\|\omega_i\|\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t)+\vartheta_i\dot{b}_i(t))\right\| \\ &= -k_{i_1}\|\omega_i\|^2 - k_{i_2}\|\omega_i\| + (y_i+D_i+k_{i_2})\|\omega_i\| + (y_i+D_i+k_{i_2})e^{-(\lambda_1+\lambda_2 y_i)t} \\ &+ \left(m_{i2}\left\|\vartheta_i\dot{p}_i - (\ddot{b}_i(t)+\vartheta_i\dot{b}_i(t))\right\| + \frac{1}{2}c_{M_i}\|\dot{p}_i\|\|\omega_i\| + D_i\right)\|\omega_i\| \\ &= -k_{i_1}\|\omega_i\|^2 - k_{i_2}\|\omega_i\| + (y_i+D_i+k_{i_2})e^{-(\lambda_1+\lambda_2 y_i)t} \\ &\leq -k_{i_1}\|\omega_i\|^2 - k_{i_2}\|\omega_i\| + y_i e^{-\lambda_2 y_i t}e^{-\lambda_1 t} + D_i e^{-(\lambda_1+\lambda_2 y_i)t} + k_{i_2}e^{-(\lambda_1+\lambda_2 y_i)t}\end{aligned}$$

From the above analysis, it can be seen that the derivative of Lyapunov function has an upper bound,

$$\dot{V}_i \leq -k_{i_1}\|\omega_i\|^2 - k_{i_2}\|\omega_i\| + y_i e^{-\lambda_2 y_i t}e^{-\lambda_1 t} + (D_i+k_{i_2})\,e^{-\lambda_1 t} \tag{4.2.11}$$

If we integrate equation (4.2.11) in the time interval $[0,+\infty)$, we obtain $\omega_i \in \mathcal{L}_1 \cap \mathcal{L}_2$. This hold true, hence $y_i e^{-\lambda_2 y_i t}e^{-\lambda_1 t} \in \mathcal{L}_1$. So, we obtain that $V_i \in \mathcal{L}_\infty$.

$$\begin{aligned}\dot{V}_i &\leq -k_{i_1}\|\omega_i\|^2 - k_{i_2}\|\omega_i\| + y_i e^{-\lambda_2 y_i t}e^{-\lambda_1 t} + (D_i+k_{i_2})\,e^{-\lambda_1 t} \\ \int_0^\infty \dot{V}_i dt &\leq -k_{i_1}\int_0^\infty \|\omega_i\|^2 dt - k_{i_2}\int_0^\infty \|\omega_i\| dt + \int_0^\infty y_i e^{-\lambda_2 y_i t}e^{-\lambda_1 t} dt \\ &+ (D_i+k_{i_2})\int_0^\infty e^{-\lambda_1 t} dt\end{aligned}$$

$$V_{i,\infty} - V_i(0) \leq -k_{i_1}\int_0^\infty \|\omega_i\|^2 dt - k_{i_2}\int_0^\infty \|\omega_i\| dt + \frac{D_i + k_{i_2}}{\lambda_1} + \int_0^\infty y_i e^{-\lambda_2 y_i t} e^{-\lambda_1 t} dt$$

$$k_{i_1}\int_0^\infty \|\omega_i\|^2 dt + k_{i_2}\int_0^\infty \|\omega_i\| dt \leq (V_i(0) - V_{i,\infty}) + \frac{D_i + k_{i_2}}{\lambda_1} + \int_0^\infty \|y_i e^{-\lambda_2 y_i t} e^{-\lambda_1 t}\| dt$$

$$k_{i_1}\int_0^\infty \|\omega_i\|^2 dt + k_{i_2} + \int_0^\infty \|\omega_i\| dt \leq (V_i(0) - V_{i,\infty}) + \frac{D_i + k_{i_2}}{\lambda_1} + \frac{1}{e\lambda_1\lambda_2}$$

Because the Lyapunov function is positive defined and radially unbounded with its derivative be bounded and negative, then the function is decreasing overtime with lower bound V_∞, such that $\lim_{t\to\infty} V_i(t) = V_\infty$, for every agent $i \in \mathcal{V}$. Thus, it has been proved that $\dot{V}_i \leq 0$. Additionally, it has been proven that the sliding surface variables are bounded, $\omega_i \in \mathcal{L}_\infty$. Therefore, it can be proven that the error variables can be bounded. As the function is bounded and the sliding surface variables are also bounded, then from equation (4.2.1) can be proved that the NOCGPROX variables are also bounded. So,

$$\begin{aligned}
\dot{q}_i &= -\vartheta_i q_i + \omega_i \\
q_i(t) &= e^{-\vartheta_i t} q_i(0) + \int_0^t e^{-\vartheta_i (t-s)} \omega_i(s) ds \\
\|q_i(t)\| &\leq e^{-\vartheta_i t}\|q_i(0)\| + \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds
\end{aligned}$$

hence $\|\omega_i(s)\| \leq c_\omega$ then the above inequality will have form,

$$\begin{aligned}
\|q_i(t)\| &\leq e^{-\vartheta_i t}\|q_i(0)\| + c_\omega \int_0^t e^{-\vartheta_i (t-s)} ds \\
&= e^{-\vartheta_i t}\|q_i(0)\| + \frac{c_\omega}{\vartheta_i}(1 - e^{-\vartheta_i t}) \\
&\leq c_q := \max\left\{\|q_i(0)\|, \frac{c_\omega}{\vartheta_i}\right\}
\end{aligned}$$

So, the variable $q_i \in \mathcal{L}_\infty$. The next step is to seek the boundedness of the derivative. From the equation (4.2.1) it is

$$\dot{q}_i = -\vartheta_i q_i + \omega_i \Rightarrow \|\dot{q}_i\| \leq \vartheta_i \|q_i\| + \|\omega_i\| \Rightarrow \|\dot{q}_i\| \leq c_{\dot{q}} := \vartheta_i c_q + c_\omega$$

Thus the derivative $\dot{q}_i \in \mathcal{L}_\infty$. In addition, this implies that all the other signals are bounded. Thus, it has been proven that $\dot{p}_i, \dot{b}_i, \ddot{b}_i \in \mathcal{L}_\infty$. Taking into account the boundedness of these signals, along with the NOCGPROX variables, it can be seen that $p_i, b_i \in \mathcal{L}_\infty$. Therefore, from $b_i(t)$ definition, $s_i(\lfloor \frac{t}{T} \rfloor T) \in \ell_\infty$, hence $\rho(t)$ is bounded by definition. As a result, it has been proven that the sampled measurements of the neighbor agents are bounded $x_j(kT) \in \ell_\infty$. The next step of the stability analysis is to ensure that the error variables are square integrable, in order to ensure that they converge to zero. As it has been proved an

upper bound for the Lyapunov derivative, it can be utilized to ensure the necessary condition. So, from equation (4.2.1) it can be shown that the variables $q_i \in \mathcal{L}_2$.

$$\begin{aligned}
\dot{q}_i &= -\vartheta_i q_i + \omega_i \\
q_i(t) &= e^{-\vartheta_i t} q_i(0) + \int_0^t e^{-\vartheta_i (t-s)} \omega_i(s) ds \\
\|q_i(t)\| &\leq e^{-\vartheta_i t} \|q_i(0)\| + \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds \\
\|q_i(t)\|^2 &\leq \left(e^{-\vartheta_i t} \|q_i(0)\| + \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds \right)^2
\end{aligned}$$

Using Hölder's inequality in the above expression, it can be shown that

$$\|q_i(t)\|^2 \leq e^{-2\vartheta_i t} \|q_i(0)\|^2 + \int_0^t e^{-2\vartheta_i (t-s)} \|\omega_i(s)\|^2 ds$$

Now integrate both parts of the inequality in the interval $[0, \infty)$ it can be seen that

$$\begin{aligned}
\int_0^\infty \|q_i(t)\|^2 dt &\leq \int_0^\infty e^{-2\vartheta_i t} \|q_i(0)\|^2 dt + \int_0^\infty \int_0^t e^{-2\vartheta_i (t-s)} \|\omega_i(s)\|^2 ds dt \\
&= \frac{\|q_i(0)\|^2}{2\vartheta_i} + \int_0^\infty \int_0^t e^{-2\vartheta_i (t-s)} \|\omega_i(s)\|^2 ds dt
\end{aligned}$$

It has been proven that the sliding surface variables are $\omega_i \in \mathcal{L}_2$. So, utilizing Fubini's theorem, the double integral can be computed. Nonetheless, it must be ensured that both integral spaces are measurable. It can be seen that both independent integrals are bounded, and their respected spaces are measurable over time. So, the variable t will be set in the interval $[s, \infty]$ and variable s in the interval $[0, \infty]$. Thus,

$$\begin{aligned}
\int_0^\infty \|q_i(t)\|^2 dt &\leq \frac{\|q_i(0)\|^2}{2\vartheta_i} + \int_0^\infty \|\omega_i(s)\|^2 \left(\int_s^\infty e^{-2\vartheta_i (t-s)} dt \right) ds \\
&= \frac{\|q_i(0)\|^2}{2\vartheta_i} + \frac{1}{2\vartheta_i} \underbrace{\int_0^\infty \|\omega_i(s)\|^2 ds}_{<\infty}
\end{aligned}$$

Therefore, it has been proven that the variables $q_i \in \mathcal{L}_2$ with bound,

$$\int_0^\infty \|q_i(t)\|^2 dt \leq \frac{\|q_i(0)\|^2}{2\vartheta_i} + \frac{1}{2\vartheta_i} \int_0^\infty \|\omega_i(s)\|^2 ds < \infty \tag{4.2.12}$$

Since it has been proved that $q_i \in \mathcal{L}_\infty \cap \mathcal{L}_2$ and $\dot{q}_i \in \mathcal{L}_\infty$, then from Barbalát's lemma is shown that the $\lim_{t\to\infty} q_i(t) = 0$. Therefore, the necessary condition for the proposed variables has been achieved. Furthermore, because the error variables tend to zero from the regulation problem, then its samples will also converge to zero as the iteration of the algorithm increases. Thus, $\lim_{\nu\to\infty} q_i^{\nu D} = 0$, and the algorithm will converge to the desired

stationary point. The last step is to prove that the error variables are also $\mathcal{L}_1$. From equation (4.2.1) it can be shown

$$\begin{aligned}
\dot{q}_i &= -\vartheta_i q_i + \omega_i \\
q_i(t) &= e^{-\vartheta_i t} q_i(0) + \int_0^t e^{-\vartheta_i (t-s)} \omega_i(s) ds \\
\|q_i(t)\| &\leq e^{-\vartheta_i t} \|q_i(0)\| + \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds \\
\int_0^\infty \|q_i(t)\| dt &\leq \int_0^\infty e^{-\vartheta_i t} \|q_i(0)\| dt + \int_0^\infty \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds dt \\
&\leq \frac{\|q_i(0)\|}{\vartheta_i} + \int_0^\infty \int_0^t e^{-\vartheta_i (t-s)} \|\omega_i(s)\| ds dt
\end{aligned}$$

In order to have the $\mathcal{L}_1$ integral of ω_i in the above the expression, Fubini's theorem will be used to transform the planes of the integrals. Nonetheless, it must be ensured that both integral spaces are measurable. It can be seen that both independent integrals are bounded, and their respected spaces are measurable over time. So, the variable t will be set in the interval $[s, \infty]$ and variable s in the interval $[0, \infty]$. Thus,

$$\begin{aligned}
\int_0^\infty \|q_i(t)\| dt &\leq \frac{\|q_i(0)\|}{\vartheta_i} + \int_0^\infty \|\omega_i(s)\| \left(\int_s^\infty e^{-\vartheta_i (t-s)} dt \right) ds \\
&= \frac{\|q_i(0)\|}{\vartheta_i} + \frac{1}{\vartheta_i} \underbrace{\int_0^\infty \|\omega_i(s)\| ds}_{<\infty}
\end{aligned}$$

So, it has been proven that $q_i \in \mathcal{L}_1$, with bound as expressed in expression (4.2.12)

$$\int_0^\infty \|q_i(t)\| dt \leq \frac{\|q_i(0)\|}{\vartheta_i} + \frac{1}{\vartheta_i} \underbrace{\int_0^\infty \|\omega_i(t)\| dt}_{<\infty} < \infty \tag{4.2.13}$$

It has been proven that the error variable $q_i \in \mathcal{L}_\infty \cap \mathcal{L}_1 \cap \mathcal{L}_2$ and its derivative $\dot{q}_i \in \mathcal{L}_\infty$. These are the required conditions in order to prove that the sampled error variables are square summable. So, this concept is illustrated by the following Lemma 4.1.

Lemma 4.1. if $q_i(t) \in \mathcal{L}_\infty \cap \mathcal{L}_1 \cap \mathcal{L}_2$ and $\dot{q}_i(t) \in \mathcal{L}_\infty$ then the samples $q_i^{kT} \in \ell_2$, $k \in \mathbf{N}$,i.e.

$$\sum_{k=0}^{\infty} \left\| q_i^{kT} \right\|^2 < \infty \tag{4.2.14}$$

Proof. The proof of this lemma begins in a constructive manner. Let be the below derivative

$$\frac{d}{dt} \left((t - kT) \|q_i(t)\|^2 \right) = \|q_i(t)\|^2 + 2(t - kT) \dot{q}_i^T(t) q_i(t)$$

the next step is to integrate the above equation in the interval $(kT, (k+1)T)$.

$$\begin{aligned}
\int_{kT}^{(k+1)T} \frac{d}{dt}\left((t-kT)\|q_i(t)\|^2\right) dt &= \int_{kT}^{(k+1)T} \left((\|q_i(t)\|^2 + 2(t-kT)\dot{q}_i^T(t)q_i(t)\right) dt \\
(t-kT)\|q_i(t)\|^2\Big|_{kT}^{(k+1)T} &= \int_{kT}^{(k+1)T} \|q_i(t)\|^2 dt + 2\int_{kT}^{(k+1)T} (t-kT)\dot{q}_i^T(t)q_i(t)dt \\
T\,\|q_i((k+1)T)\|^2 &\leq \int_{kT}^{(k+1)T} \|q_i(t)\|^2\, dt + 2T\int_{kT}^{(k+1)T} \left|\dot{q}_i^T(t)q_i(t)\right| dt
\end{aligned}$$

The final step is to sum the above expression from $k = 0...\infty$. So,

$$\begin{aligned}
\sum_{k=0}^{\infty} T\,\|q_i((k+1)T)\|^2 &\leq \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} \|q_i(t)\|^2\, dt + \sum_{k=0}^{\infty} 2T\int_{kT}^{(k+1)T} |\dot{q}_i(t)q_i(t)|\, dt \\
T\sum_{k=0}^{\infty} \|q_i((k+1)T)\|^2 &\leq \int_0^{\infty} \|q_i(t)\|^2\, dt + 2T\int_0^{\infty} |\dot{q}_i(t)q_i(t)|\, dt \\
\sum_{k=0}^{\infty} \|q_i((k+1)T)\|^2 &\leq \frac{1}{T}\int_0^{\infty} \|q_i(t)\|^2\, dt + 2\int_0^{\infty} \left|\dot{q}_i^T(t)q_i(t)\right| dt
\end{aligned}$$

Since $\dot{q}_i(t) \in \mathcal{L}_\infty$ is essentially bounded, there exist a constant $M > 0$ such that $\|\dot{q}_i(t)\| \leq M$ almost everywhere. So,

$$\begin{aligned}
\sum_{k=0}^{\infty} \|q_i((k+1)T)\|^2 &\leq \frac{1}{T}\int_0^{\infty} \|q_i(t)\|^2\, dt + 2\int_0^{\infty} \left|\dot{q}_i^T(t)q_i(t)\right| dt \\
&= \frac{1}{T}\int_0^{\infty} \|q_i(t)\|^2\, dt + 2\int_0^{\infty} \|\dot{q}_i(t)\|\|q_i(t)\| dt \\
&\leq \frac{1}{T}\int_0^{\infty} \|q_i(t)\|^2\, dt + 2\int_0^{\infty} M\,\|q_i(t)\|\, dt \\
&= \frac{1}{T}\underbrace{\int_0^{\infty} \|q_i(t)\|^2\, dt}_{<\infty} + 2M\underbrace{\int_0^{\infty} \|q_i(t)\|\, dt}_{<\infty}
\end{aligned}$$

It is known that $q_i(t) \in \mathcal{L}_1 \cap \mathcal{L}_2$. Thus, the above result is finite and bounded. Subsequently,

$$\sum_{k=0}^{\infty} \|q_i((k+1)T)\|^2 \leq \frac{1}{T}\int_0^{\infty} \|q_i(t)\|^2\, dt + 2M\int_0^{\infty} |q_i(t)|\, dt < \infty$$

$$\sum_{k=0}^{\infty} \left\|q_i^{kT}\right\|^2 < \infty$$

Therefore it has been proven that $q_i^{kT} \in \ell_2$. □

As, it has been stated previously, the variables $s_i(kT) \in \ell_\infty$, and in extent $x_i(kT) \in \ell_\infty$. It is secured that $x_j(kT) \in \ell_\infty$. From Lemma 4.1 it has been shown that $q_i(kT) \in \ell_2$. So, the (3.2.18) will decrease with each passing iteration since the error term converges to zero

and the sum of all computed samples are finite. Therefore, the primal variable converges to the stationary point, which is KKT. Therefore, from theorem 3.1 it has been proven the agents achieve optimal consensus in a sublinear manner.

As a summary, it has been proven that the NOCGPROX variables are converging to zero from Barbalát Lemma and its samples are square integrable. Therefore, with the proposed error variables and the robust distributed control law (4.2.9), optimal consensus will be achieved in a sublinear manner with asymptotic stability of the system.

Chapter 5

Simulations and Results

In this chapter, the application of the proposed methodology is demonstrated through its implementation on a swarm of unmanned aerial vehicles (UAVs). Several simulations with different cost functions have been conducted to examine the agents' motion and evaluate the effectiveness of the proposed approach. These simulations verify the boundedness and square integrability of the introduced variables, ensuring the achievement of optimal consensus among the agents. All simulations were carried out using MATLAB.

5.1 Application

Consider a group of N UAVs which are shown in figure 5.1. From [68] the kinematics of a UAV $i \in \mathcal{V}$ in 3-dimentional space are described by

$$\begin{aligned} \dot{x}_i &= V_i \cos(\chi_i)\cos(\gamma_i) \\ \dot{y}_i &= V_i \sin(\chi_i)\cos(\gamma_i) \\ \dot{z}_i &= V_i \sin(\gamma_i) \end{aligned} \tag{5.1.1}$$

where (x_i, y_i, z_i) denotes the position in the inertia frame, V_i the speed of the UAV, χ_i the heading angle of the UAV and γ_i the flight path angle of the UAV. The dynamics of the $i-$th UAV are described by

$$\begin{aligned} \dot{V}_i &= (T_i - D_i)/m_i - g\sin(\gamma_i) \\ \dot{\chi}_i &= (L_i \sin(\phi_i))/(m_i V_i \cos(\gamma_i)) \\ \dot{\gamma}_i &= (L_i \cos(\phi_i) - m_i g \cos(\gamma_i))/m_i V_i \end{aligned} \tag{5.1.2}$$

where T_i is the engine thrust, L_i the vehicle lift, D_i the vehicle drag, m_i the mass, g the gravitational constant and ϕ_i the banking angle.

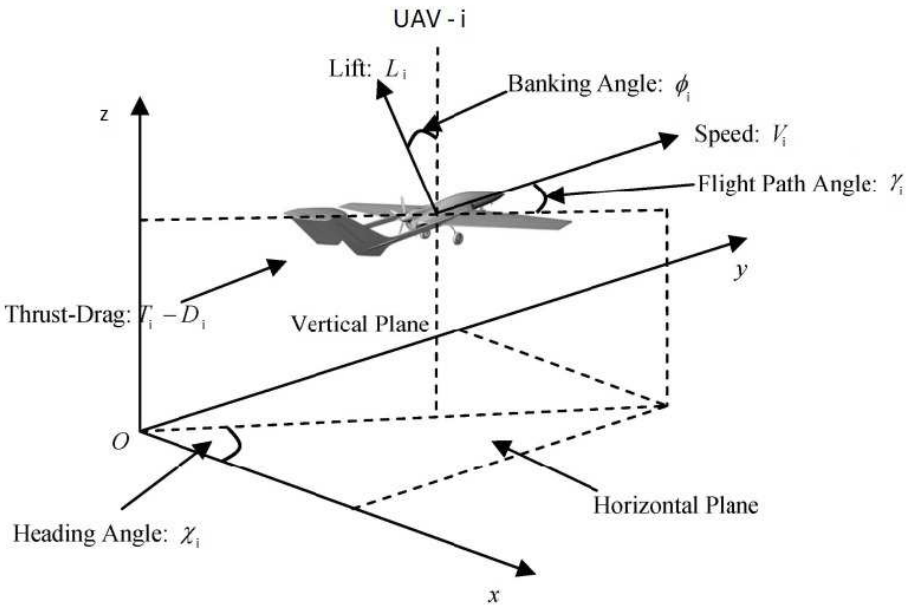

Figure 5.1: Parameters of $i-$th UAV, [68].

Let $F_i = [T_i \quad L_i \sin(\phi_i) \quad L_i \cos(\phi_i)]^T$ be the control input, $p_i = [x_i \quad y_i \quad z_i]^T$ the position vector, $v_i = \dot{p}_i = [\dot{x}_i \quad \dot{y}_i \quad \dot{z}_i]^T$ the velocity vector and $\tilde{d}_i$ an unknown bounded disturbance force vector. Then, from the equations (5.1.1), (5.1.2) it is derived that

$$\begin{aligned} \dot{p}_i &= v_i \\ m_i \dot{v}_i &= \eta_i + m_i \varepsilon_i + G_i(F_i + \tilde{d}_i) \end{aligned} \tag{5.1.3}$$

where

$$\eta_i = \begin{bmatrix} -D_i \cos(\chi_i)\cos(\gamma_i) \\ -D_i \sin(\chi_i)\cos(\gamma_i) \\ -D_i \sin(\gamma_i) \end{bmatrix} \tag{5.1.4}$$

$$\varepsilon_i = [0 \;\; 0 \;\; g]^T \tag{5.1.5}$$

and

$$G_i = \begin{bmatrix} \cos(\chi_i)\cos(\gamma_i) & -\sin(\chi_i) & -\sin(\gamma_i)\cos(\chi_i) \\ \sin(\chi_i)\cos(\gamma_i) & \cos(\chi_i) & -\sin(\chi_i)\cos(\gamma_i) \\ \sin(\gamma_i) & 0 & \cos(\gamma_i) \end{bmatrix} \tag{5.1.6}$$

It can be seen that $G_i^{-1} = G_i^T$. Every UAV communicate its position through the graph configuration in sampled time periods. Thus, its output is going to be $\boldsymbol{x}_i := \boldsymbol{p}_i$. Every UAV has a local nonconvex cost function $f_i(x_i, y_i, z_i)$. The algorithm 1 has a gradient approximation, thus only the information of the gradient is required in order to update the primal and dual variable. Consider that the input of the system is given by $F_i = G_i^T u_i$, with

$u_i = [u_{i_x} \quad u_{i_y} \quad u_{i_z}]^T$ and $d_i = G_i \tilde{d}_i$. Then, the dynamics of every agent will be:

$$\begin{aligned} \dot{p}_i &= v_i \\ m_i \dot{v}_i &= \eta_i + m_i \varepsilon_i + u_i + d_i \\ x_i &= p_i \end{aligned} \tag{5.1.7}$$

5.2 System Simulation with MATLAB

In this section, the simulations of the proposed algorithm with the control scheme will be presented. The system is a connected network of $N = 5$ UAV agents which have dynamics from (5.1.7) and control laws from (4.2.9),(4.2.10). All the signals of the closed-loop system ought to be bounded, while the error variable tends to zero, as the agents reach an optimal stationary point or a desired formation.

5.2.1 Parameters of the System

Every agent has a local cost function $f_i(x_i, y_i, z_i)$, $i \in \mathcal{V}$. The communication graph of the agents is shown in figure 5.2 . It can be seen that is connected as is perquisite of the assumption A.4.

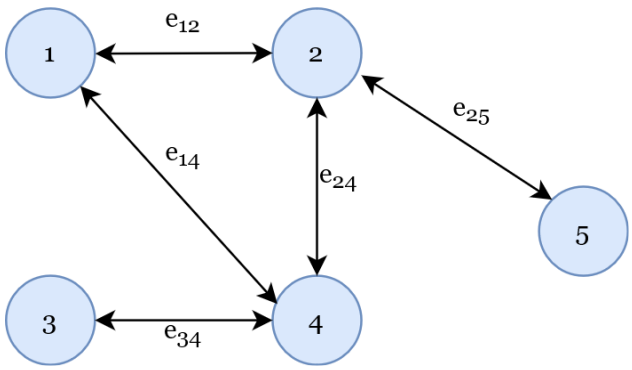

Figure 5.2: Communication Graph of UAVs.

The incidence matrix of the communication graph of figure 5.2, based on definition 2.9 with its elements computed by equation (2.1.1), is

$$\mathcal{A} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix}$$

The Laplacian matrix of the graph will be computed by equation (2.1.3), which is also

the signed Laplacian. So

$$L = L_- = \mathcal{A}^T\mathcal{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 3 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 0 \\ -1 & -1 & -1 & 3 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow L = \begin{bmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 3 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 0 \\ -1 & -1 & -1 & 3 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix}$$

The degree matrix of the graph can be computed from definitions (2.7) and (2.8). So,

$$D = diag\{2, 3, 1, 3, 1\}$$

Finally, the signless Laplacian matrix will be computed from equation (2.1.4). So,

$$L_+ = 2D - \mathcal{A}^T\mathcal{A} = \begin{bmatrix} 4 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & -1 & 0 & -1 & 0 \\ -1 & 3 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 0 \\ -1 & -1 & -1 & 3 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 3 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

The initial conditions of every $i-$th agent is presented by the below vectors.

$$x(0) = [1, \quad 1.5, \quad 4, \quad -0.1, \quad 3]^T$$
$$y(0) = [0, \quad -2, \quad 1, \quad 5, \quad 2.5]^T$$
$$z(0) = [1, \quad 2.4, \quad 2, \quad 0.5, \quad 3.4]^T$$

while the masses of the agents are presented in table 5.1.

i	1	2	3	4	5
$m_i(kg)$	2.1	3.2	2	1.8	2.8

Table 5.1: Mass of every agent.

The parameters of the system that change in every scenario are presented in table 5.2.

Parameters	Description
$x_i(0)$,$y_i(0)$,$z_i(0)$	Initial conditions of every agent
x_{obs}	Position of an obstacle
τ	Obstacle distance
T	Sampled Period Time
d_{max}	Communication delay
β	penalty parameter
c	Algorithm candidate function constant
ϵ	Algorithm convergence value
L	Lipschitz constant
ϑ_i	Sliding manifold gain parameters
k_{i_1}, k_{i_2}	Controller gains
λ_1, λ_2	Exponential gain parameters

Table 5.2: Variables of simulation

5.2.2 Scenario 1: Optimal Consensus with Obstacle Avoidance

In this scenario, each agent is assigned a distinct local cost function designed to enable obstacle avoidance. A suitable non-convex function for each agent is defined as

$$f_i(x_i) = \frac{1}{4}\left\|x_i - x_i(0)\right\|^4 - \frac{\tau}{2}\left\|x_i - x_{obs}\right\|^2 + \frac{\tau}{1 + \left\|x_i - x_{obs}\right\|^2}$$

with $\tau > 0$. This cost function encourages each agent to maintain proximity to its initial position $x_i(0)$ while simultaneously avoiding the static obstacle located at x_{obs}. The design promotes energy-efficient motion during the simulation, as discussed in [10].

The parameter τ controls the effective influence of the obstacle. Specifically, when $\tau = 0$, no obstacle is present, while large values of τ correspond to obstacles located farther from the agents' initial positions. In this particular simulation, the distance parameter is set to $\tau = 10$, and the obstacle coordinates are defined as $x_{obs} = [-1.0,\ 0.2,\ 1.5]^T$. The system parameters are chosen as follows: $c = 10$, $\beta = 60$, which were computed from (3.2.21), (3.2.22), and $d_{max} = 0.1$, $T = 1\ sec$, $k_{i_1} = 300$, $k_{i_2} = 1$ $\lambda_1 = 0.01$, $\lambda_2 = 0.05$ and $\vartheta_i = 0.2$. The theoretical optimal point was determined using MATLAB's optimization solver and is given by:

$$x^* = \begin{bmatrix} 4.23 \\ 2.61 \\ 2.19 \end{bmatrix}$$

From Figures 5.3, 5.4, and 5.5, it can be observed that the agents successfully converge to the optimal point, demonstrating that optimal consensus has been achieved. During the simulation, the agents effectively avoid the obstacle while progressing toward their designated

goal positions. The results clearly show convergence to the theoretical optimum described earlier.

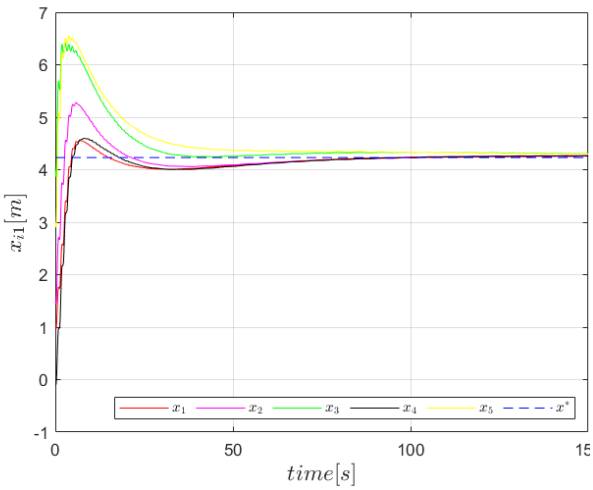

Figure 5.3: Output trajectory x_{i1} of UAVs with respect to time.

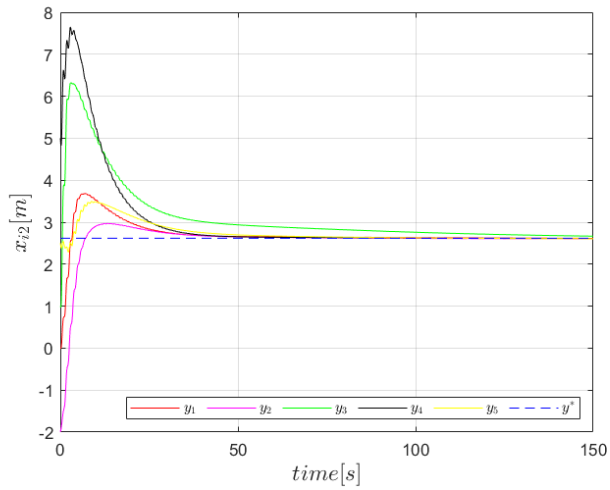

Figure 5.4: Output trajectory x_{i2} of UAVs with respect to time.

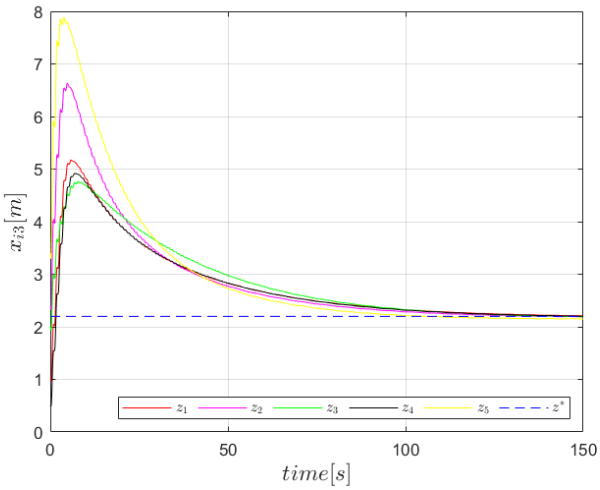

Figure 5.5: Output trajectory x_{i3} of UAVs with respect to time.

In addition to the response of the output vector, the corresponding control signals that generate this behavior are illustrated in Figures 5.6, 5.7, and 5.8. For the x and y coordinates, the control inputs are observed to oscillate around zero, suitably counteracting the sinusoidal disturbances and ensuring that each agent maintains its desired position in the Cartesian plane. In the z coordinate, the control signal oscillates around g, enabling the agents to stabilize at the desired altitude. After a brief transient period, the controller successfully drives the agents' outputs to their target positions, thereby achieving consensus.

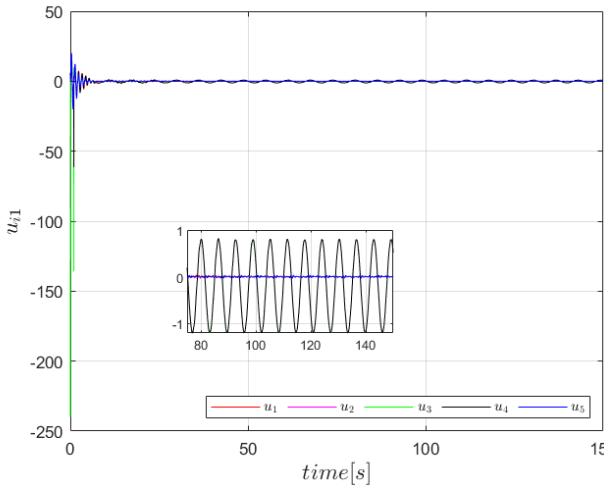

Figure 5.6: Control input u_{i1} of UAVs with respect to time.

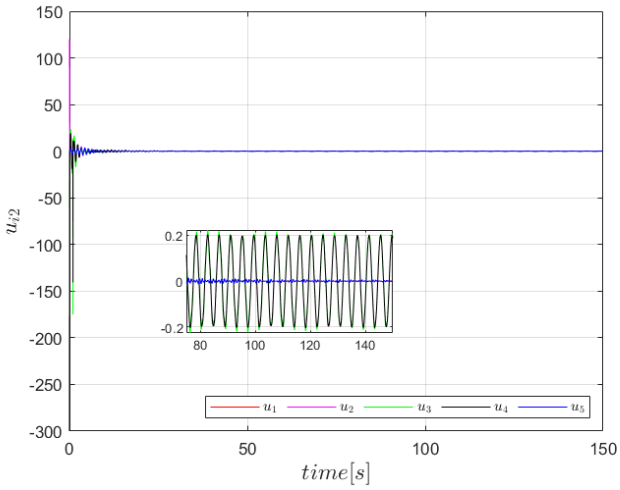

Figure 5.7: Control input u_{i2} of UAVs with respect to time.

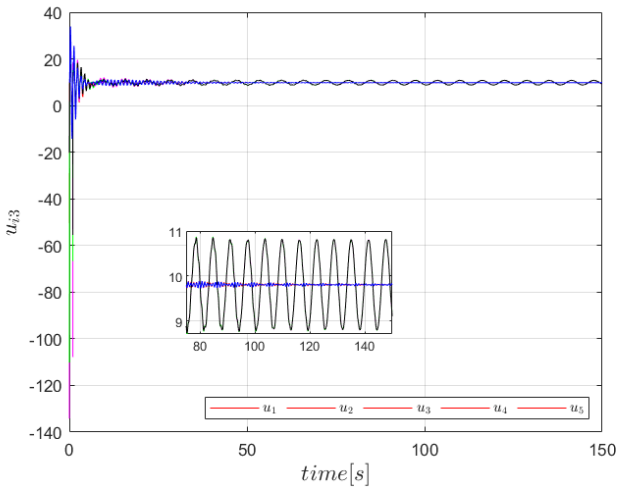

Figure 5.8: Control input u_{i3} of UAVs with respect to time.

In order to achieve consensus, the regulation variables must converge to zero. This behavior is clearly illustrated in Figure 5.9, which confirms the validity of the theoretical analysis.

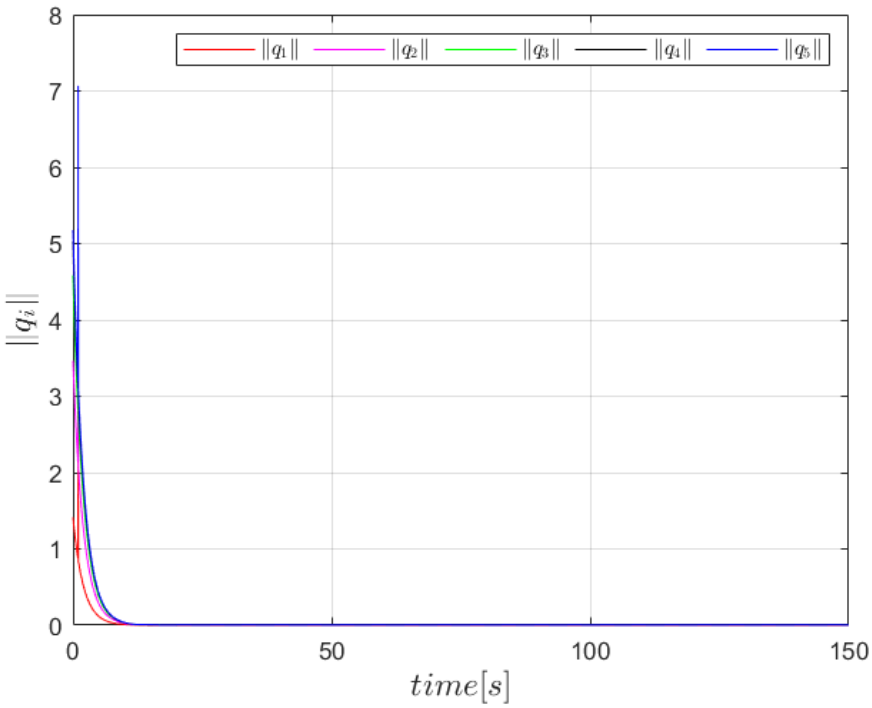

Figure 5.9: NOCGPROX variables $\|q_i\|$ with respect to time.

5.2.3 Scenario 2: Formation Control with Obstacle Avoidance

In addition to the optimal consensus problem, the agents are also required to perform formation control tasks. In this context, the agents converge to distinct target positions that form a predefined geometric pattern around an optimal solution, while simultaneously avoiding a static obstacle located at x_{obs}. This problem is known as the optimal formation control problem, which can be reformulated as an optimal consensus problem by introducing the transformed variables

$$x_{f_i} = \begin{bmatrix} R\cos\left(\frac{\pi}{3}i\right) \\ R\sin\left(\frac{\pi}{3}i\right) \\ z_f \end{bmatrix}, i = 1, ..., 5 \tag{5.2.1}$$

for a radius $R > 0$. The new optimal consensus problem for the transformed output variables $\bar{x}_i$ is formulated with the cost function $g(\bar{x}_i) = f_i(\bar{x}_i + x_{f_i})$, expressed as:

$$g(\bar{x}_i) = \frac{1}{4}\|\bar{x}_i + x_{f_i} - x_i(0)\|^4 - \frac{\tau}{2}\|\bar{x}_i + x_{f_i} - x_{obs}\|^2 + \frac{\tau}{1 + \|\bar{x}_i + x_{f_i} - x_{obs}\|^2}$$

where $\tau > 0$. The corresponding NOCGPROX variables are defined as:

$$q_i(t) = \bar{x}_i(t) - \rho\left(t - \left\lfloor\frac{t}{T}\right\rfloor T\right)\bar{s}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right) - \left[1 - \rho\left(t - \left\lfloor\frac{t}{T}\right\rfloor T\right)\right]\bar{s}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T - T\right)$$

where the discrete time vector s_i is given by:

$$\bar{s}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right)=\frac{1}{2}\bar{x}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right)+\frac{1}{2\beta d_i}\left[\beta\sum_{j\in\mathcal{N}(i)}\bar{x}_j\left(\left\lfloor\frac{t}{T}\right\rfloor T\right)-\nabla g_i\left(\bar{x}_i\left(\left\lfloor\frac{t}{T}\right\rfloor T\right)\right)\right.$$
$$\left.-\beta\sum_{j\in\mathcal{N}(i)}\sum_{\ell=1}^{\lfloor t/DT\rfloor}\left(\bar{x}_i\left(\ell T\right)-\bar{x}_j\left(\ell T\right)\right)\right]$$

for every agent $i\in\mathcal{V}$, with $\beta>0$. The control laws maintain the same structure as in the previous scenario and ensure that the error variable samples converge to zero while remaining square summable. Consequently, the swarm achieves the desired formation around the optimal configuration. In this specific simulation scenario, the parameters are chosen as follows: $\tau=10$, $x_{obs}=[-1.0,\ \ 0.2,\ \ 1.5]^T$, $c=10$, $\beta=60$ (computed from (3.2.21), (3.2.22)). Additional parameters are set to $d_{max}=0.1$, $T=1\ sec$, $k_{i_1}=300$, $k_{i_2}=1$ $\lambda_1=0.01$, $\lambda_2=0.05$ and $\vartheta_i=0.2$. In figure (5.10) we observe the final formation of the agents. As shown in Figure 5.10, the agents successfully converge to their designated formation points, confirming the achievement of optimal formation control.

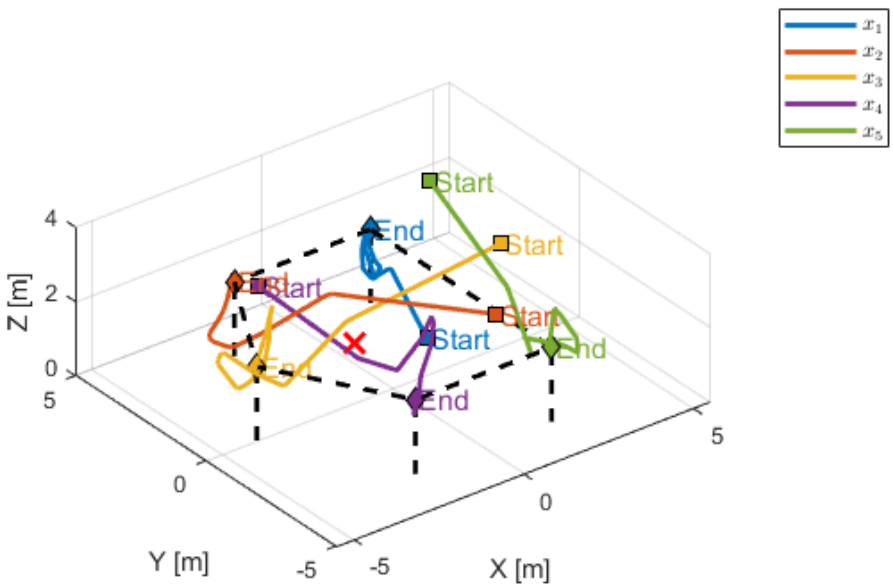

Figure 5.10: Formation control of the agents.

The trajectories of every agent in their respective coordinates can be seen in the figures (5.11), (5.12) and (5.13). As it can be seen from these figures, the agents converge in their desired position with a small margin error. The reference positions for each agent are determined from the point set defined in (5.2.1). It is evident that in all coordinate axes, the agents converge toward their optimal positions, thereby achieving optimal formation and forming a pentagonal structure.

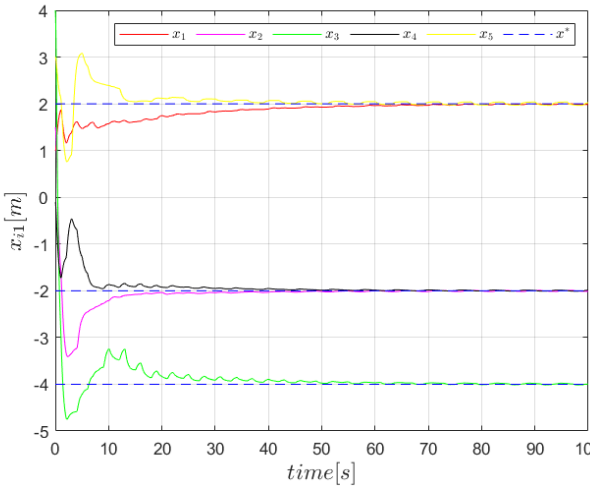

Figure 5.11: Output trajectory x_{i1} of UAVs with respect to time.

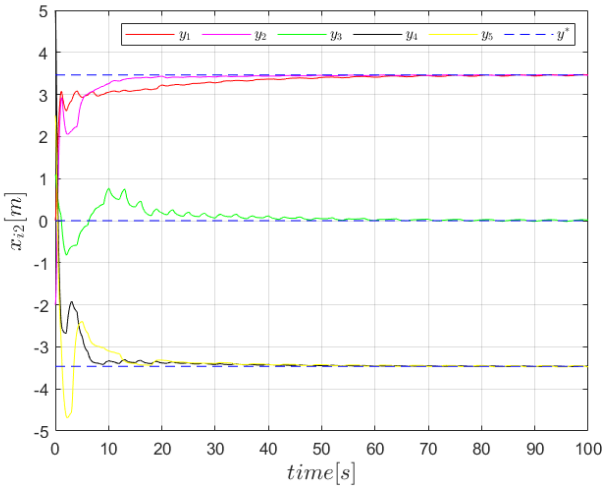

Figure 5.12: Output trajectory x_{i2} of UAVs with respect to time.

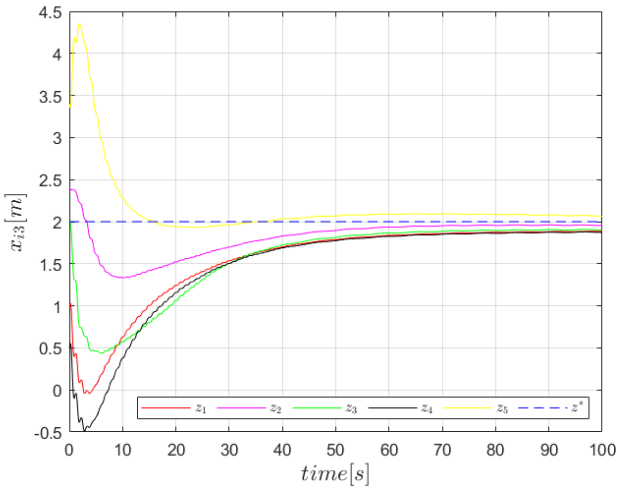

Figure 5.13: Output trajectory x_{i3} of UAVs with respect to time.

To achieve the desired formation, appropriate control inputs are generated for each agent. These control signals are depicted in Figures 5.14, 5.15, and 5.16. Similar to the optimal consensus case, the control inputs exhibit comparable dynamic behavior. For the x and y coordinates, the control signals oscillate around zero after a finite number of iterations, maintaining the desired planar position. In the z coordinate, the inputs oscillate around g, enabling the agents to stabilize at the target altitude. Although achieving formation requires greater control effort—and thus higher energy consumption—compared to the consensus

case, the generated input signals effectively guide the agents to the desired configuration, thereby ensuring successful formation control.

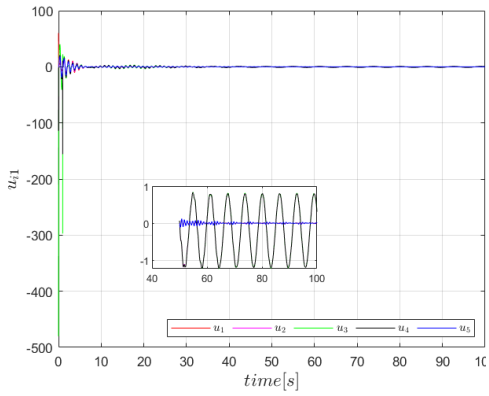

Figure 5.14: Control input u_{i1} ofUAVs with respect to time.

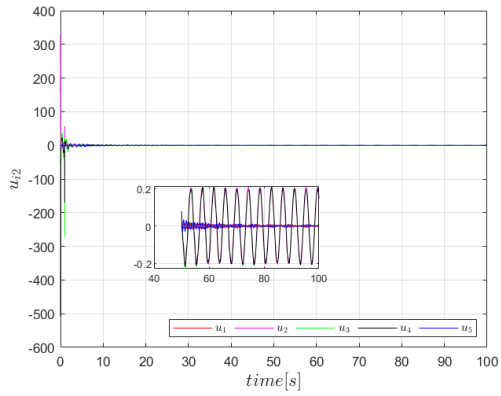

Figure 5.15: Control input u_{i2} of UAVs with respect to time.

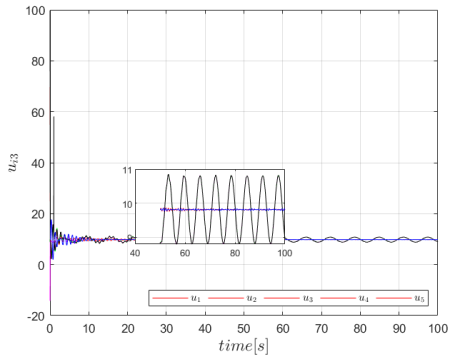

Figure 5.16: Control input u_{i3} of UAVs with respect to time.

Lastly, in figure (5.17) the NOCGPROX variables are shown. After a certain period of time, the variables converge to zero. Thus, the agents achieve the desired formation. Thus, the system is bounded and converges to zero. Consequantly, asymptotic stability has been achieved.

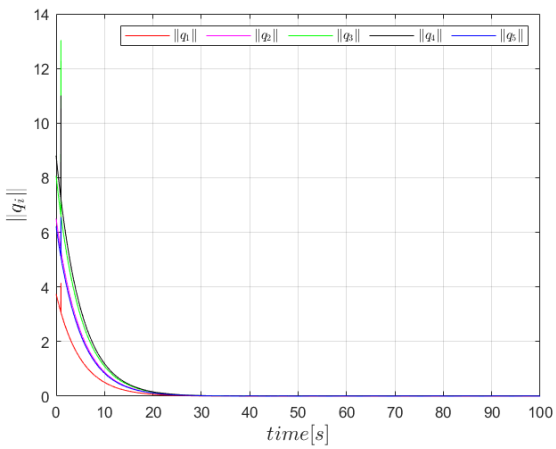

Figure 5.17: NOCGPROX variables $\|q_i\|$ with respect to time.

Chapter 6

Epilogue

6.1 Conclusions

The primary objective of this thesis was to address the proximal optimal consensus problem for a class of nonlinear agents characterized by non-convex cost functions. Inspired by the Prox-GPDA algorithm [4], we introduced a new set of error variables—termed the Non-Convex Gradient Proximal (NOCGPROX) variables. These variables employ continuously sampled output measurements to reformulate the optimal consensus problem into a regulation problem, facilitating a tractable control-based solution.

To achieve optimal consensus among the agents, we developed robust distributed control laws based on a sliding-mode-inspired feedback linearization approach. These control laws incorporate smooth approximations of sign-like functions, enhancing robustness and reducing chattering effects.

A key contribution of this work lies in establishing the ℓ_2 property for the sampled error variable sequence, which ensures the algorithm's stability and convergence. This property, together with the proposed control framework, guarantees bounded closed-loop signals and asymptotic stability of the overall system. Moreover, the algorithm demonstrates sublinear convergence with a limited number of iterations, highlighting its computational efficiency and practical applicability.

6.2 Future Work

The methodology proposed in this thesis assumes full-state measurements. However, in many practical applications, sensors provide only partial observations of the state vector. Future research can therefore explore output feedback control schemes that operate without direct velocity measurements.

Another promising direction is the extension of the framework to directed and possibly switching communication topologies, which more accurately represent realistic multi-agent networks.

Furthermore, while the current approach ensures obstacle avoidance at the final configuration through appropriate cost function design, Control Barrier Functions (CBFs) could be integrated to guarantee continuous safety throughout the agents' motion. This enhancement would enable reliable and collision-free navigation, allowing agents to reach consensus optimally while maintaining safe distances from obstacles.

Finally, the proposed method can be extended to adversarial network scenarios involving malicious agents. In such settings, mechanisms must be developed to detect and mitigate the influence of adversarial information, ensuring robust convergence to the true optimal solution.

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